

Building Molecular Maps using CellDesigner

Computational Systems Biology for Digital Medicine

7–12 December 2025

Wellcome Genome Campus, UK

Instructor: Anna Niarakis

Network construction

What is CellDesigner?

CellDesigner is a structured diagram editor for drawing gene-regulatory and biochemical networks that uses standard formats.

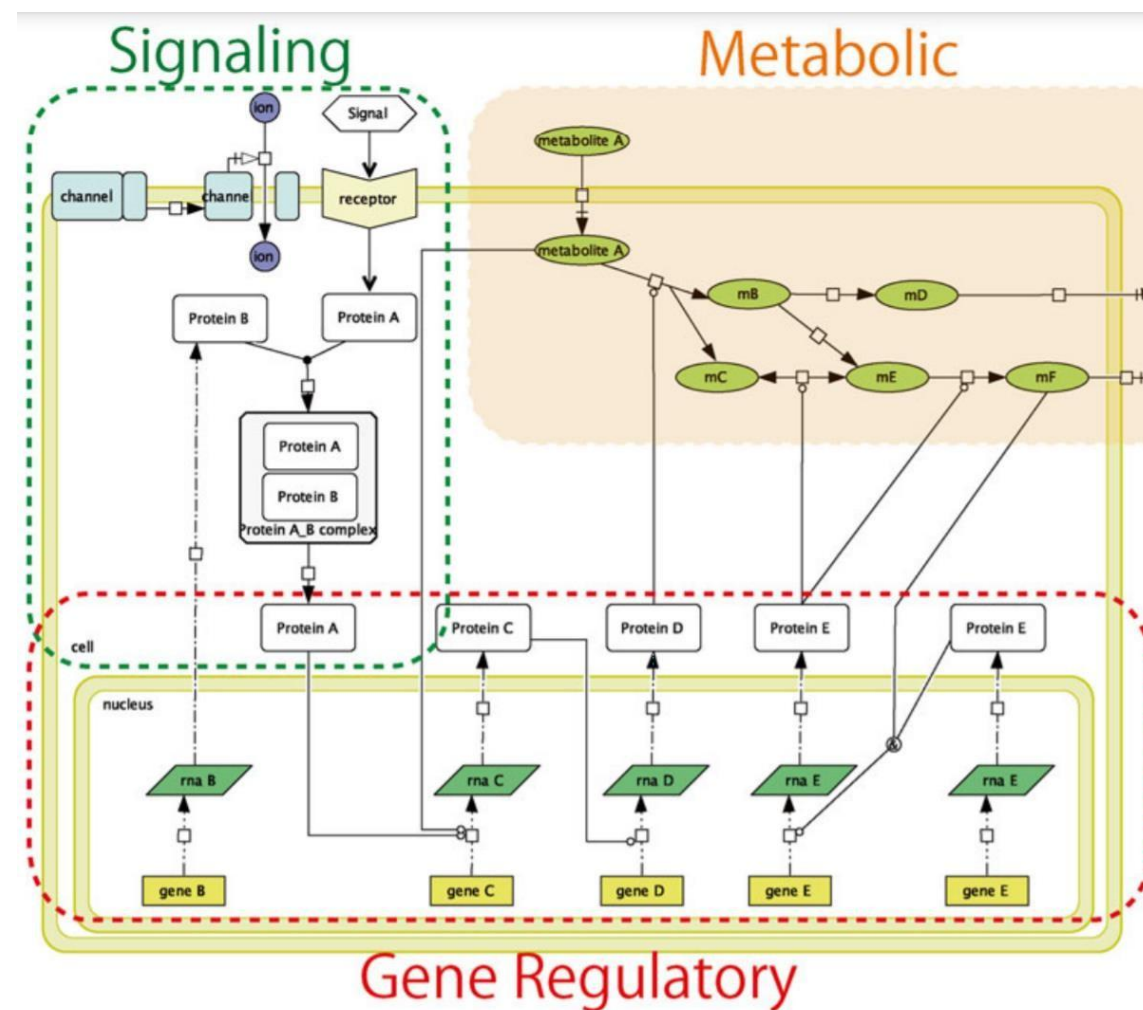


Fig. 1 Signaling/gene regulatory/metabolic pathways captured in a model created by CellDesigner

CellDesigner content



Full SBML support



Graphical notation (SBGN)



Possibility for mathematical simulations



Database connection



Export to PDF, PNG, ...



Freely available



Supported Environment

Windows

Mac OS X

Linux

What is CellDesigner good for?



CellDesigner:



Offers an « easy-to-use » graphical representation of a network



Facilitates exchange of models developed by different groups, using a standard language (SBML: Systems Biology Markup Language)



Allows to annotate the reactions, proteins, genes, etc.



Proposes links to other databases (PubMed, IHOP, BioModels, KEGG, etc.)



Organizes a lot of information in a unique diagram



Allows an exchange with other modeling tools

Install and Start CellDesigner

Download CellDesigner version 4.2.2
from <http://celldesigner.org>



General view

TREE AREA

displays all the list of the components in a tree structure.

DRAW AREA

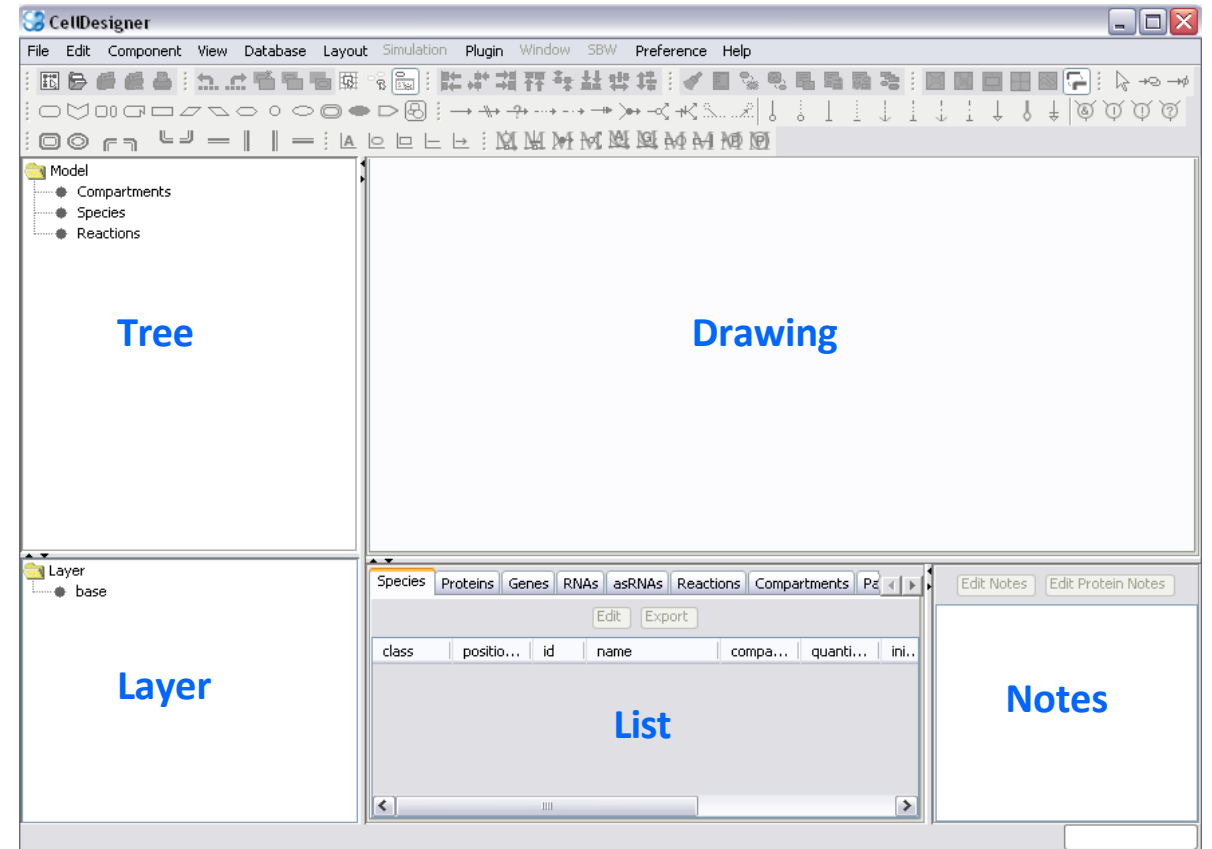
is the area where the model is built from the tool bar items.

LIST AREA

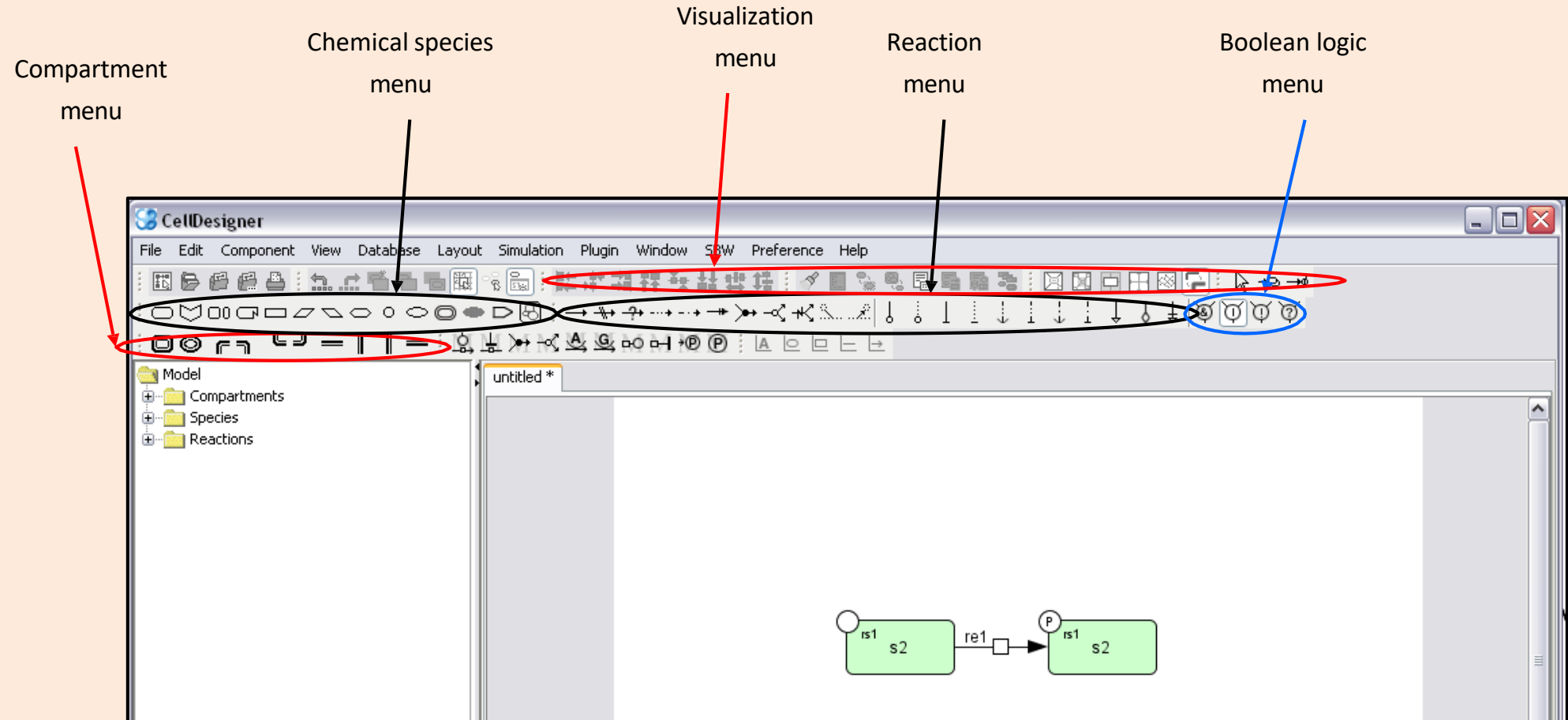
displays and edits the list of the components of the model (species, proteins, genes, RNAs, etc.)

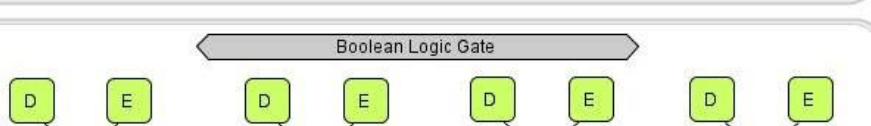
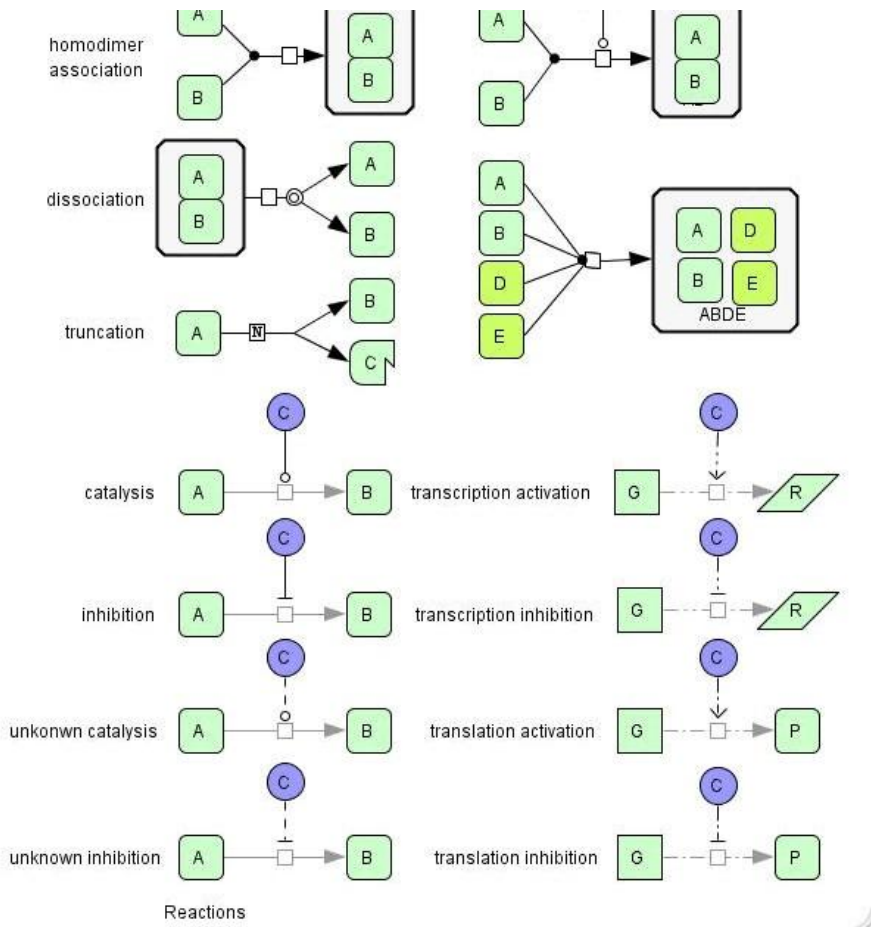
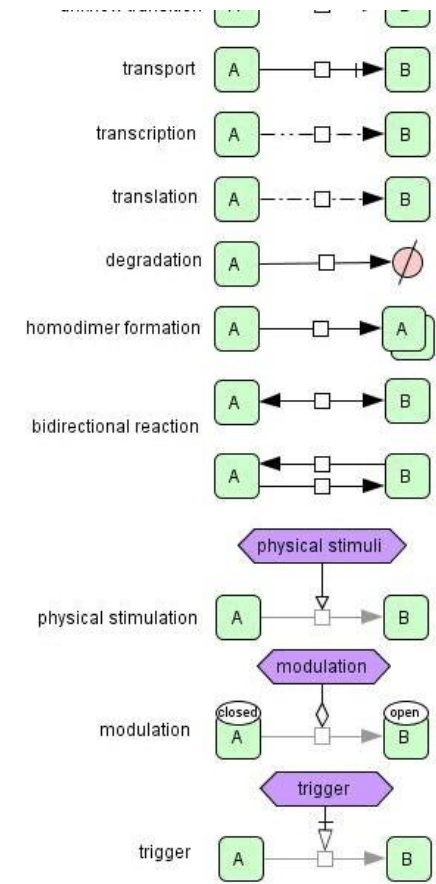
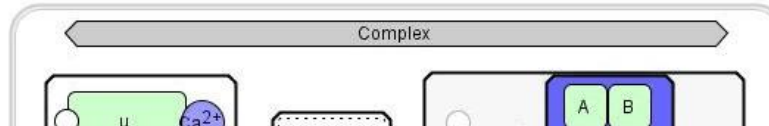
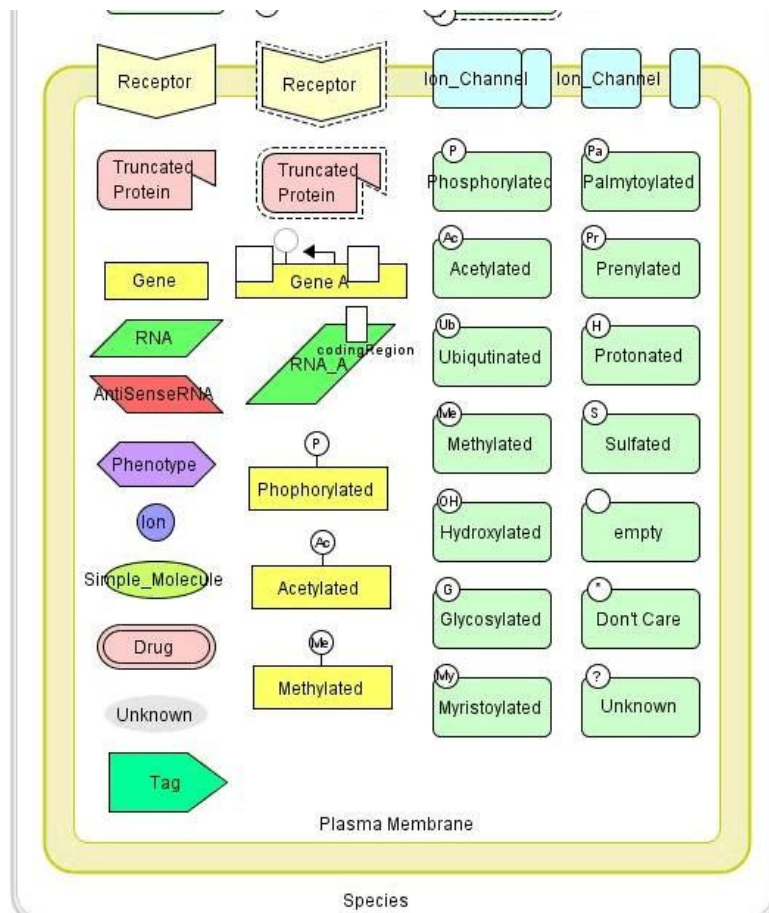
NOTES AREA

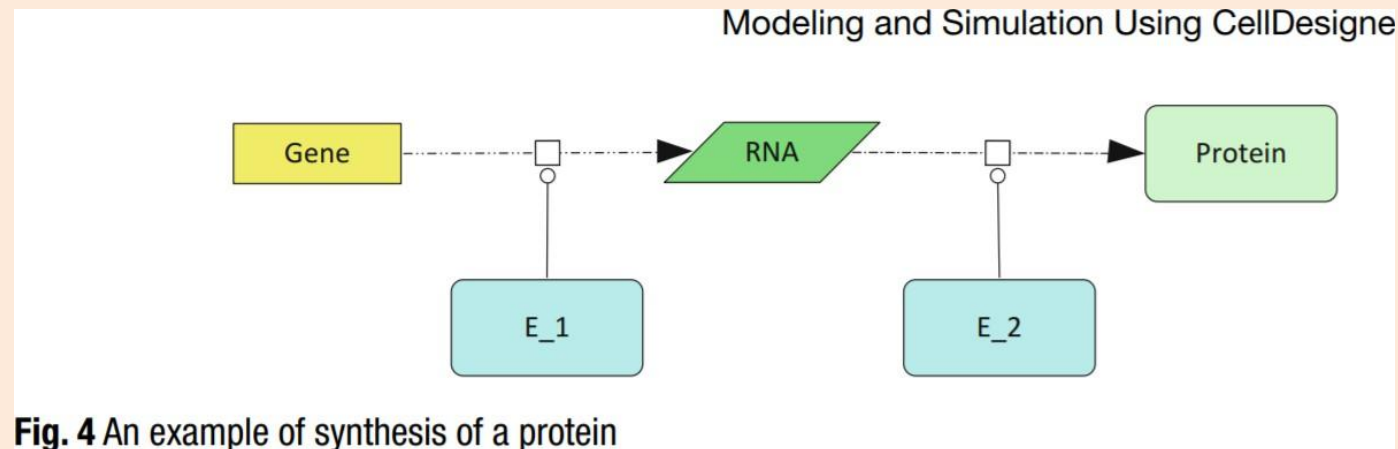
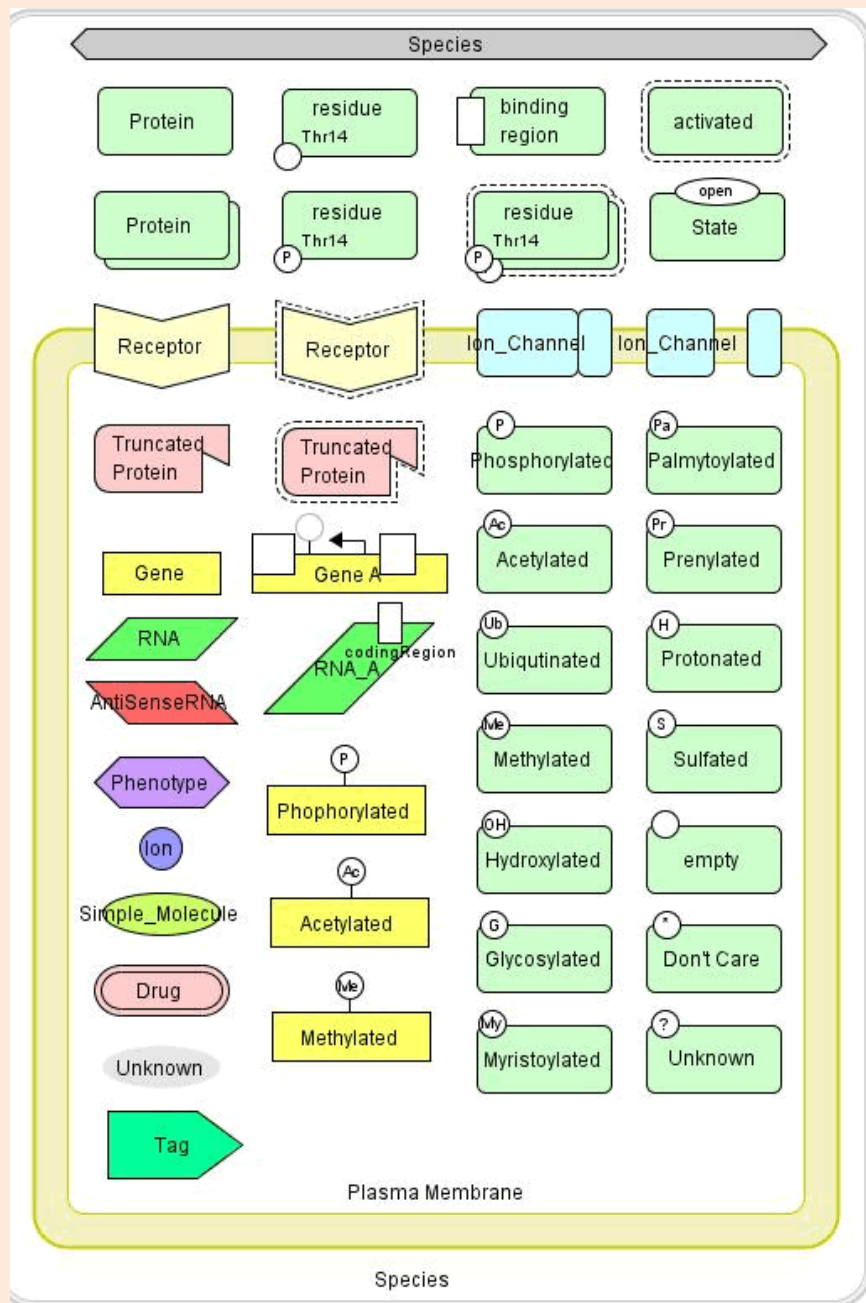
displays and edits the notes of each component (reactions, protein, complex, etc.)



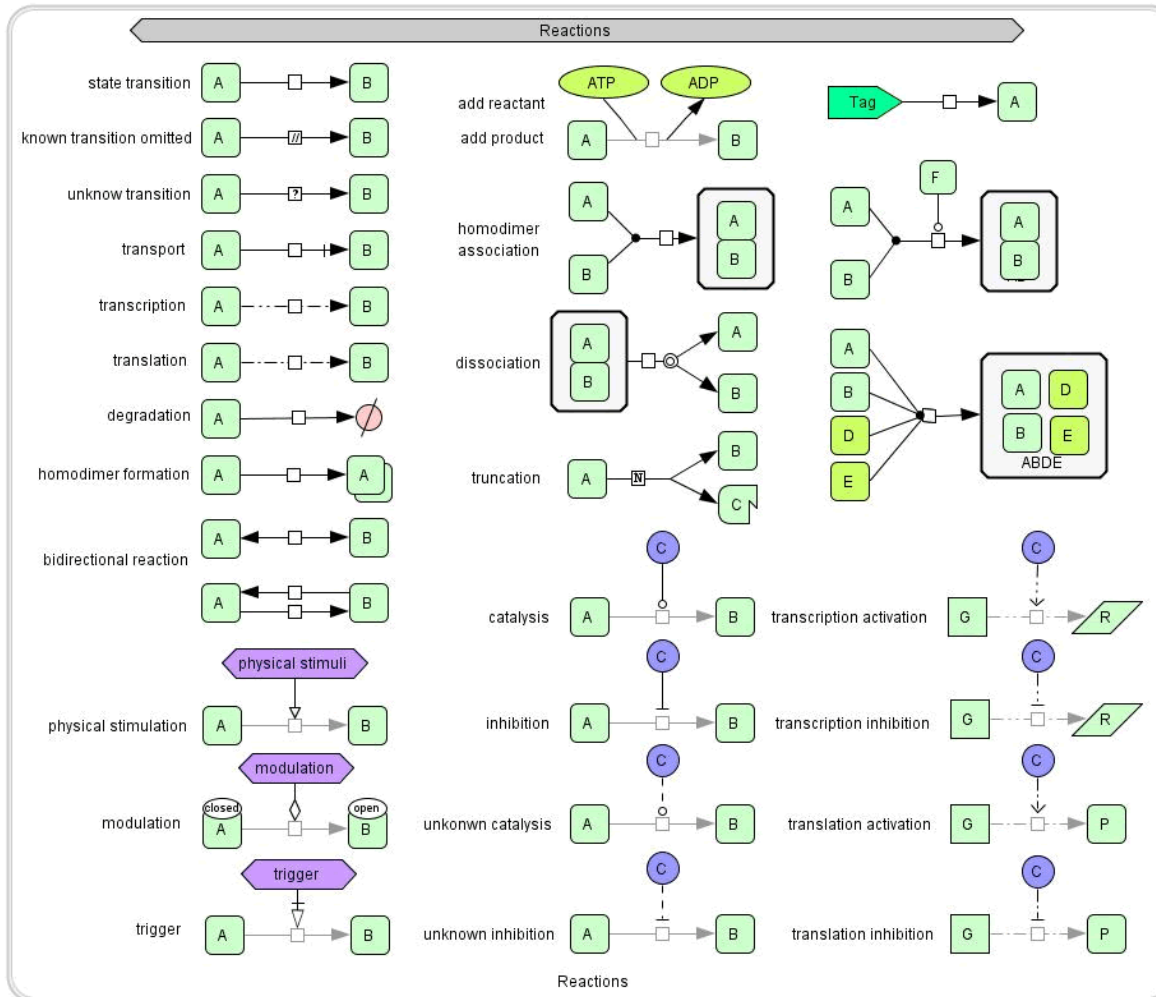
The menu







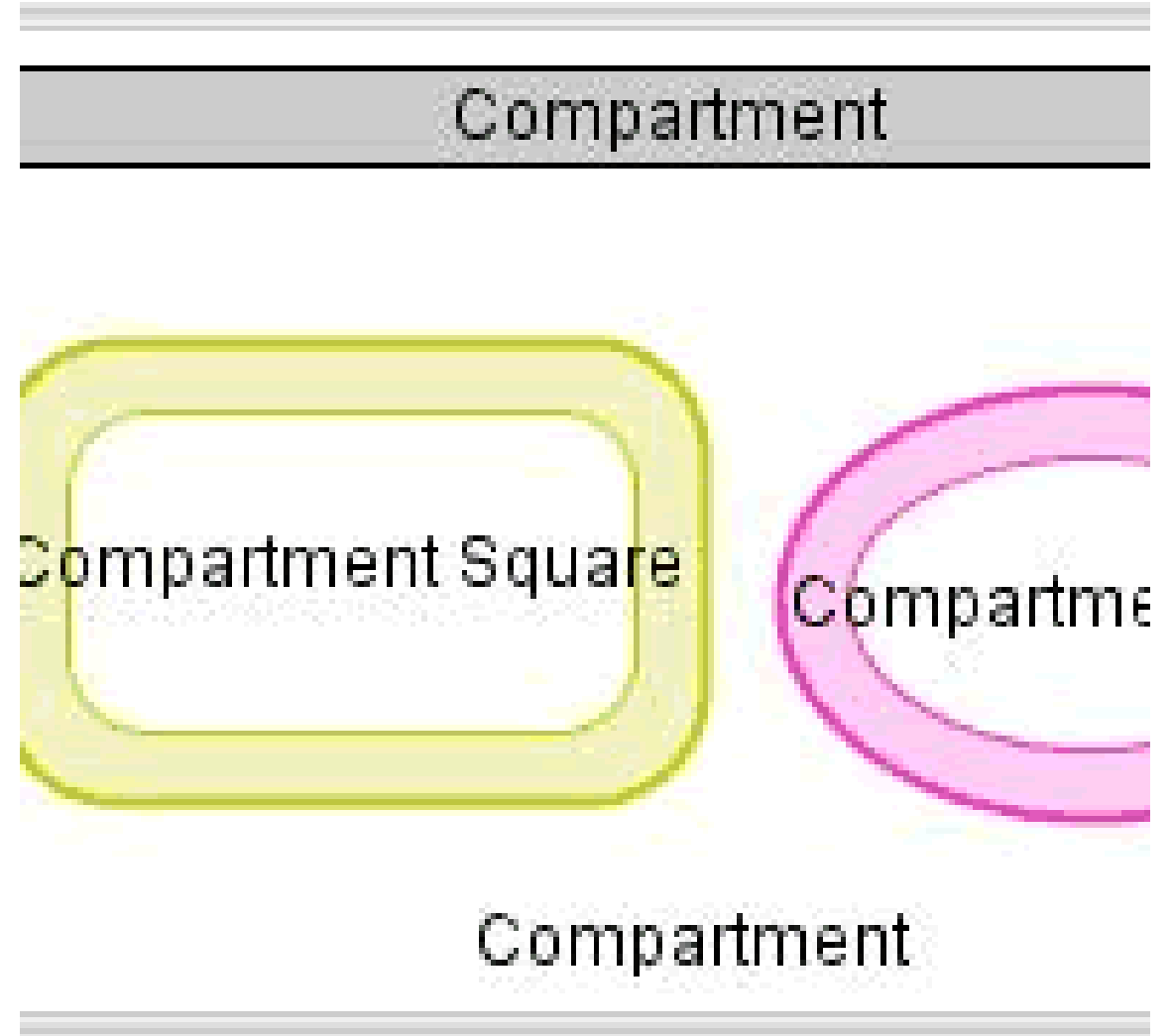
The chemical species



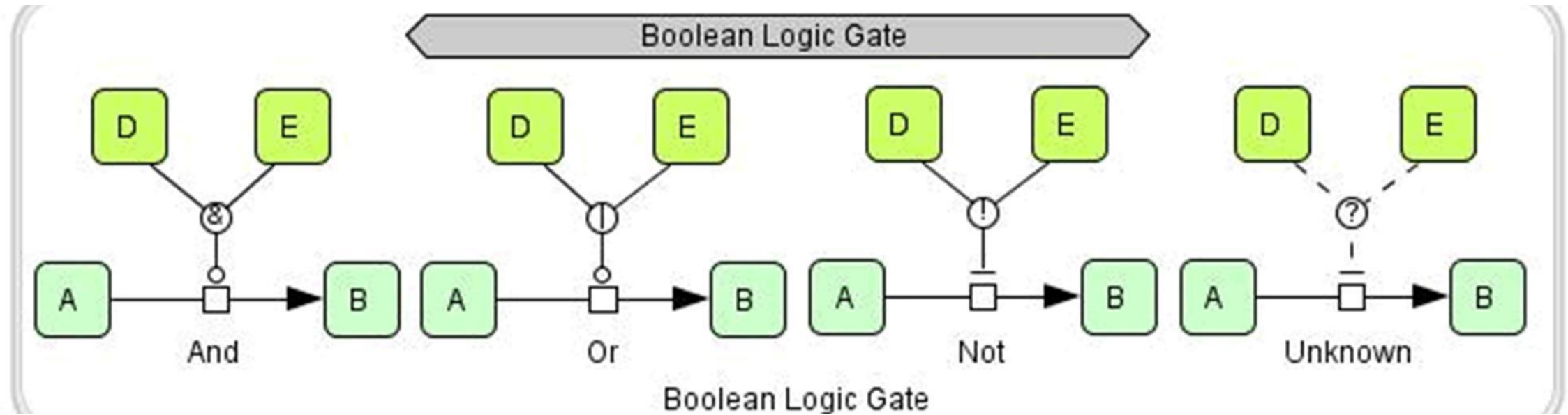
The reactions

The compartments

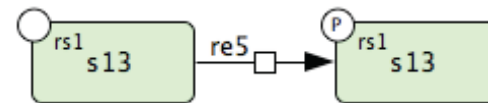
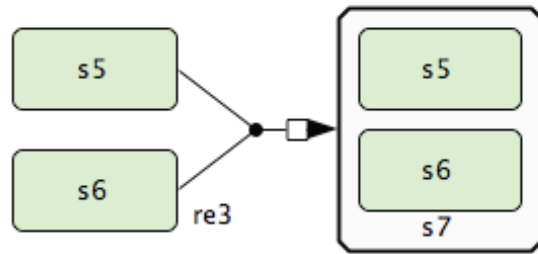
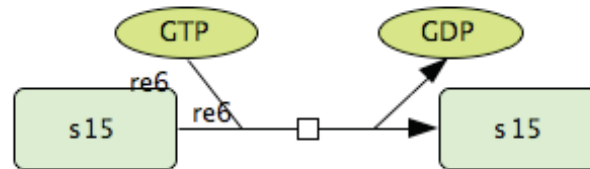
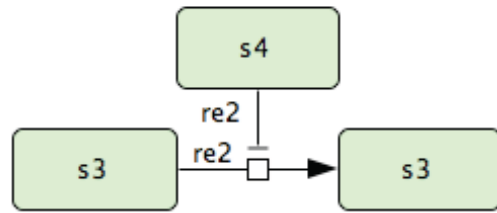
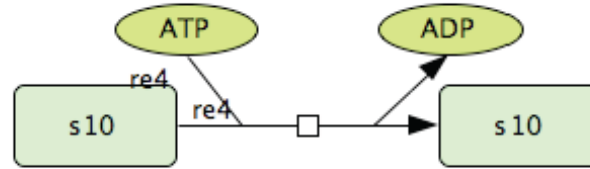
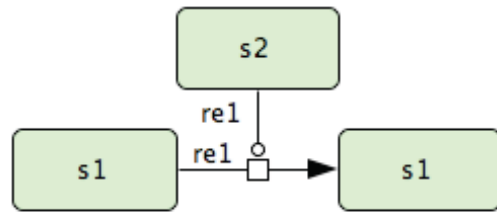
- Compartments can be:
- Cytoplasm
- Nucleus
- Nucleolus
- etc.



Boolean gates



Introduce boolean logic information into the diagram



Examples of macros

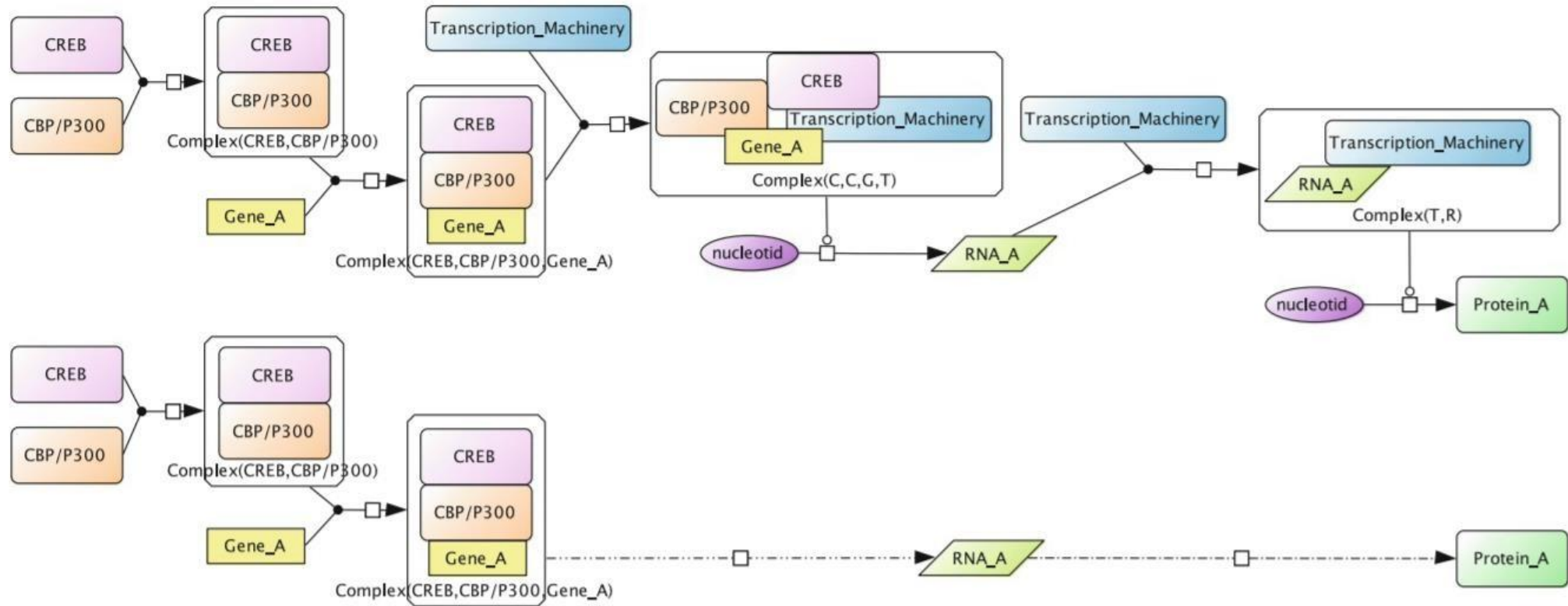


Fig. 5 An example of transcription and translation diagram (CellDesigner sample file: Sample/nbt/CD4.1/ SuplFig4a_TranscriptionTranslation_4.xml) [30]

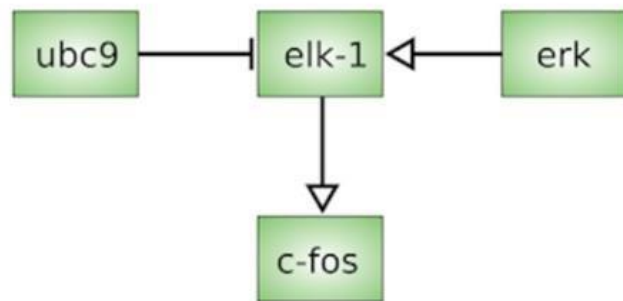


<https://sbgn.github.io/>



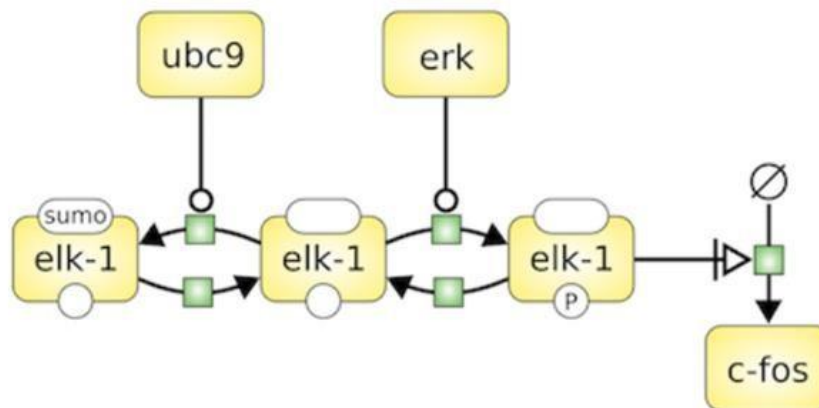
[WHAT IS SBGN](#)
[SPECIFICATIONS](#)
[SOFTWARE](#)
[EVENTS](#)
[FAQ](#)
[HOW TO CITE](#)
[ABOUT](#)
[CONTACT](#)

activity flows



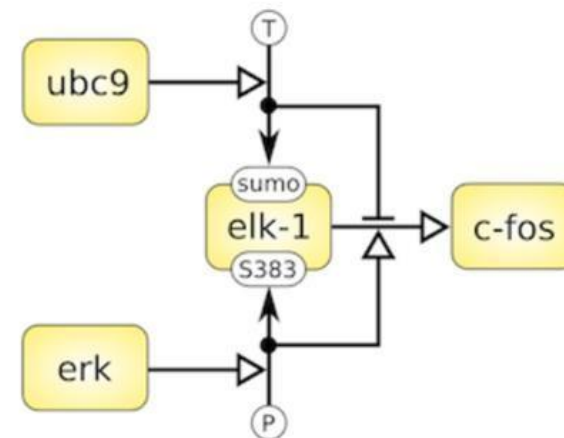
directed ☒
 sequential ☒
 mechanistic ☒

process descriptions



directed ☒
 sequential ☒
 mechanistic ☒

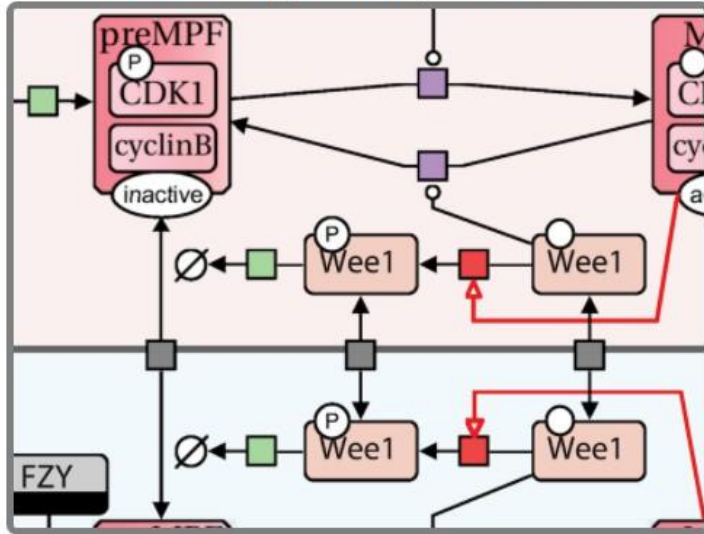
entity relationships



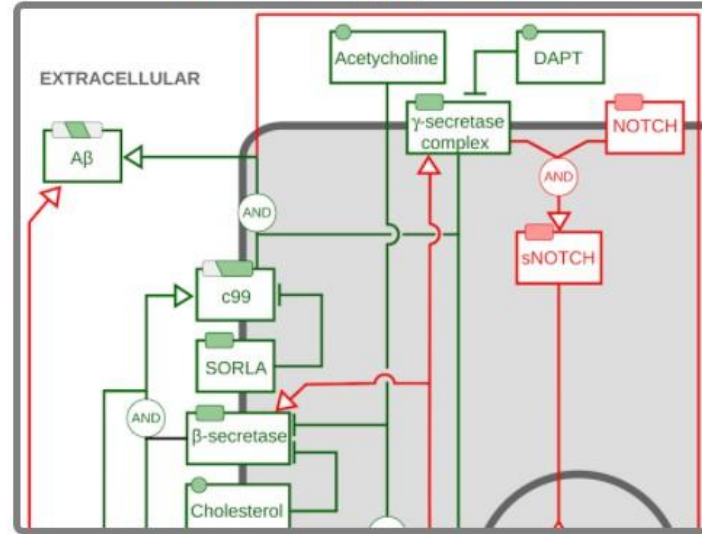
directed ☒
 sequential ☒
 mechanistic ☒

Cool visualization

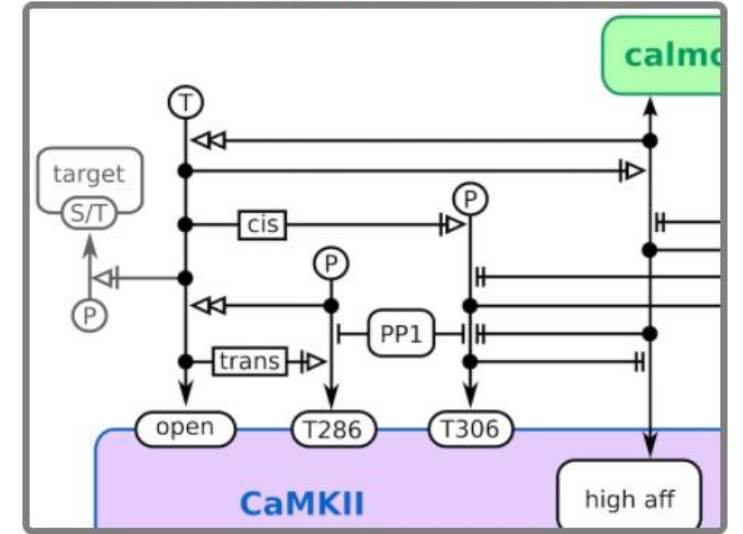
PD map of the Drosophila cell cycle,
[doi:10.1371/journal.pcbi.1005740](https://doi.org/10.1371/journal.pcbi.1005740)



AF map of protein precursor processing,
[doi:10.1002/psp4.12155](https://doi.org/10.1002/psp4.12155)



ER map of CaMKII regulation by calmodulin,
[doi:10.1371/journal.pone.0029406](https://doi.org/10.1371/journal.pone.0029406)

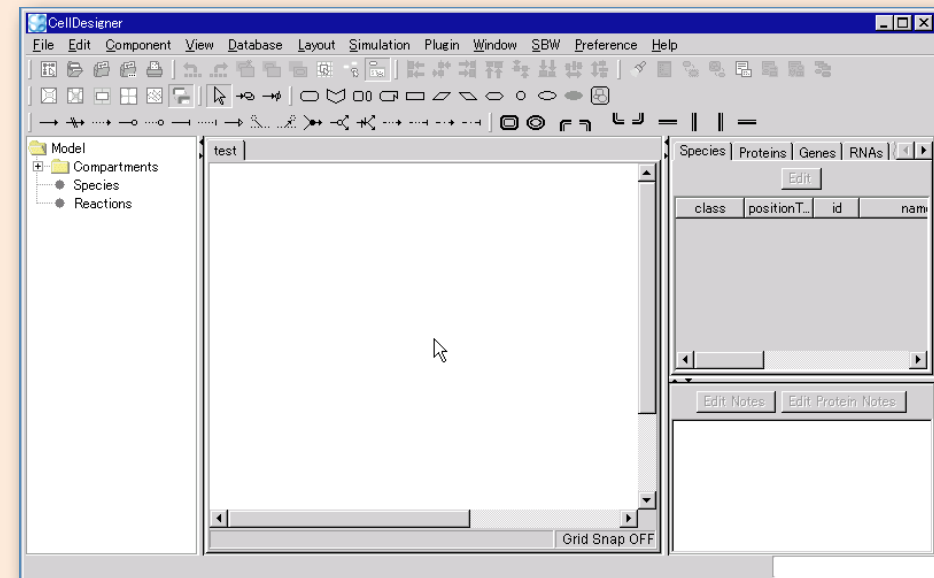
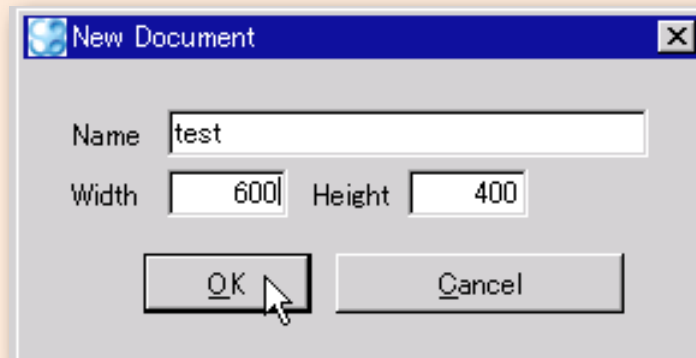
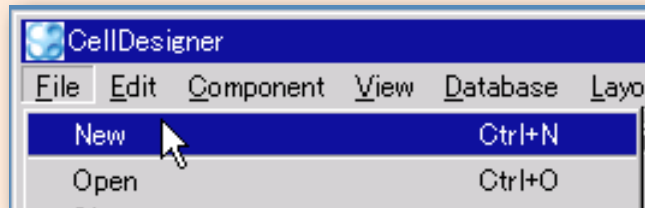


Standard notation schemes

Demonstration

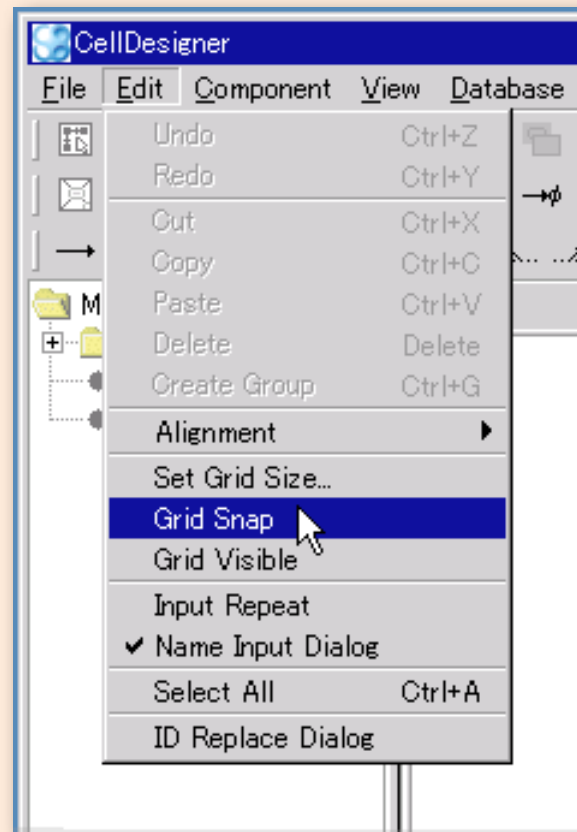
● Create new model:

● [File] → [New] → input title → [OK]



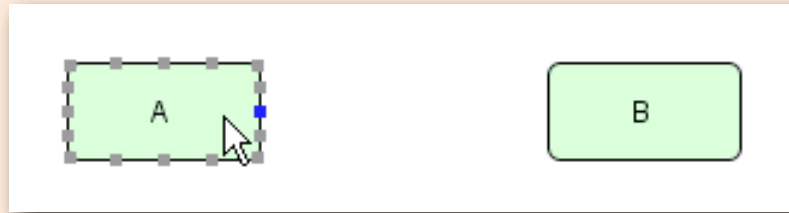
Tips

- Enable [Grid Snap] will help you draw your model much easier



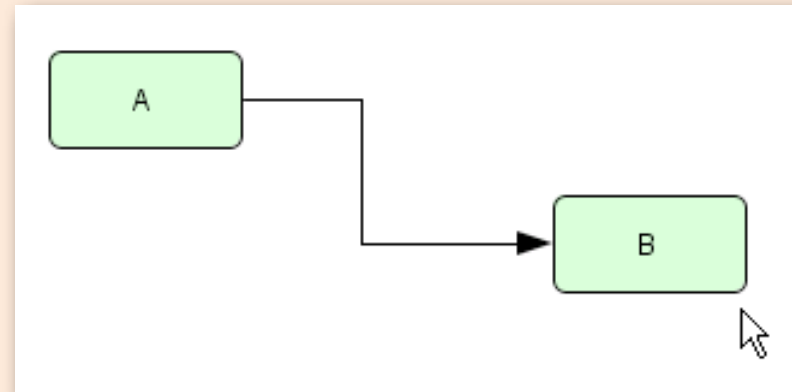
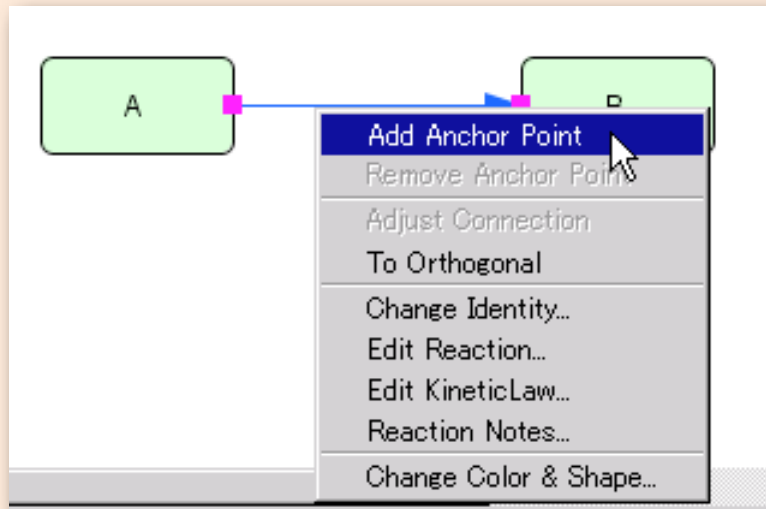
Create Reaction

- Create Protein “A” and “B”
- Draw “State transition” arrow from “A” to “B”



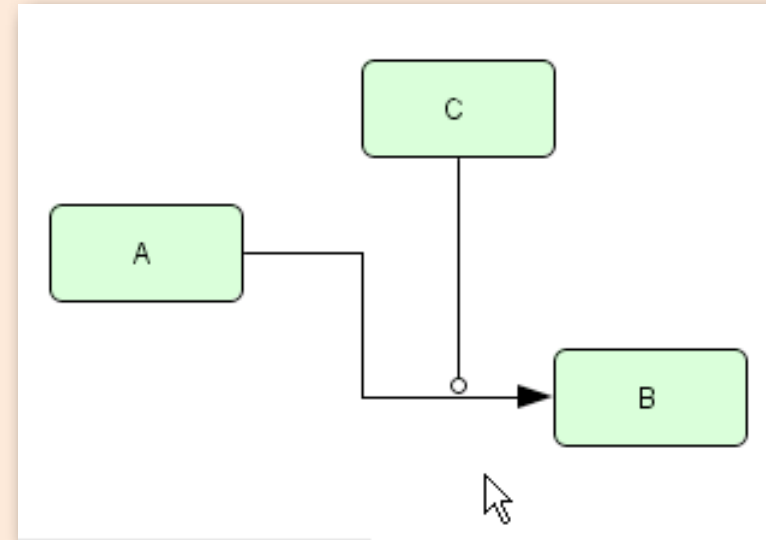
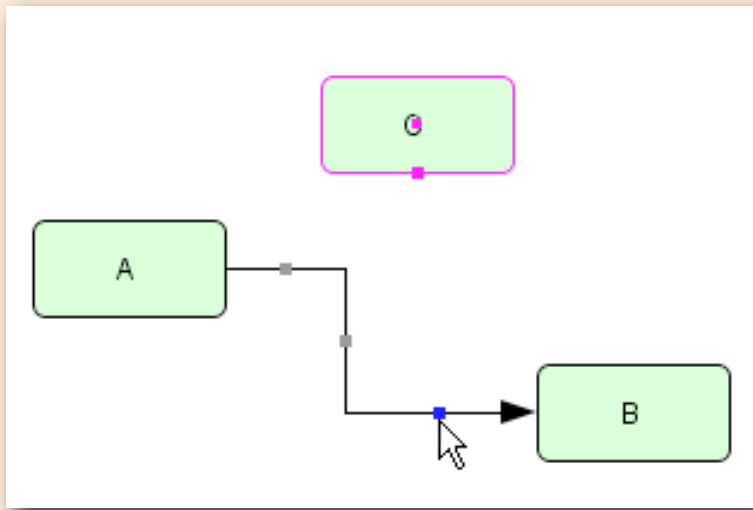
Add Anchor Point

- Add 2 anchor points to reaction
- Drag reaction and anchor point to change its shape



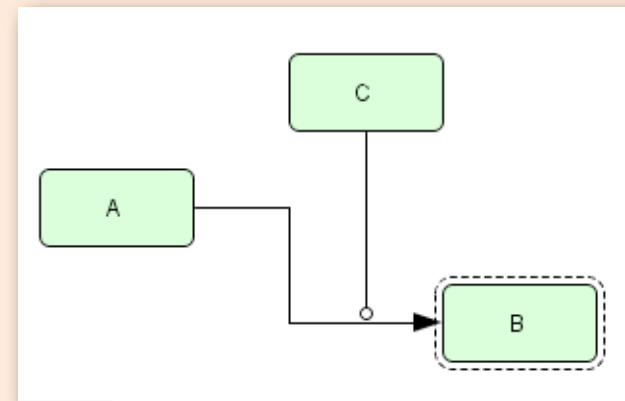
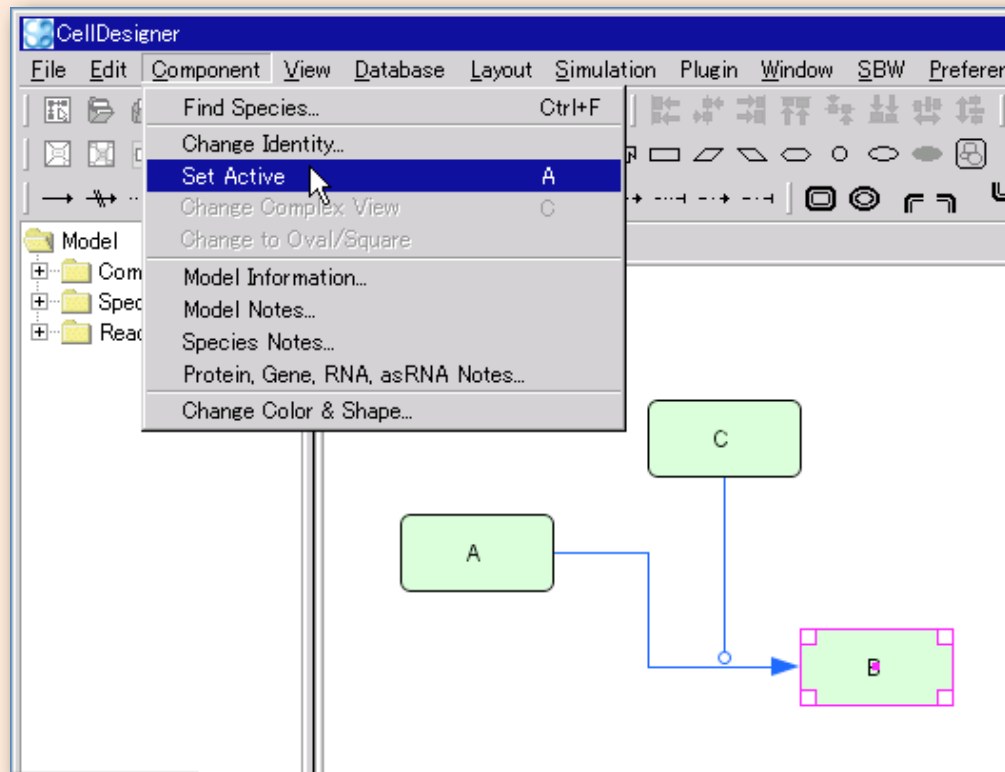
Add Catalysis reaction

- Add Protein “C”
- Add Catalysis reaction from “C” to the reaction



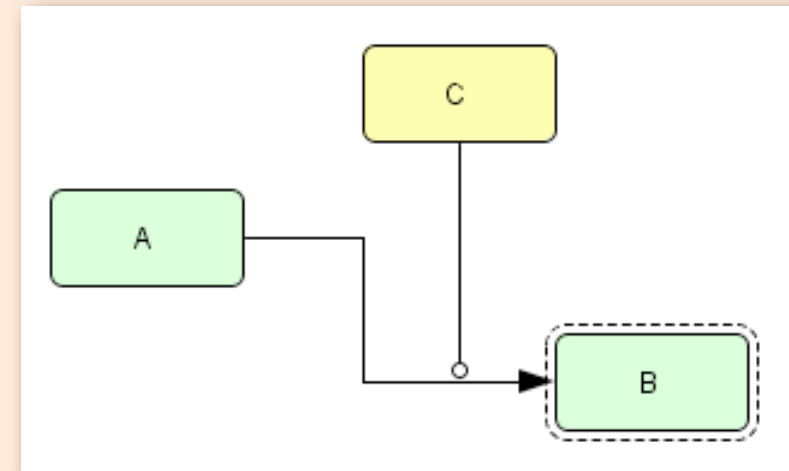
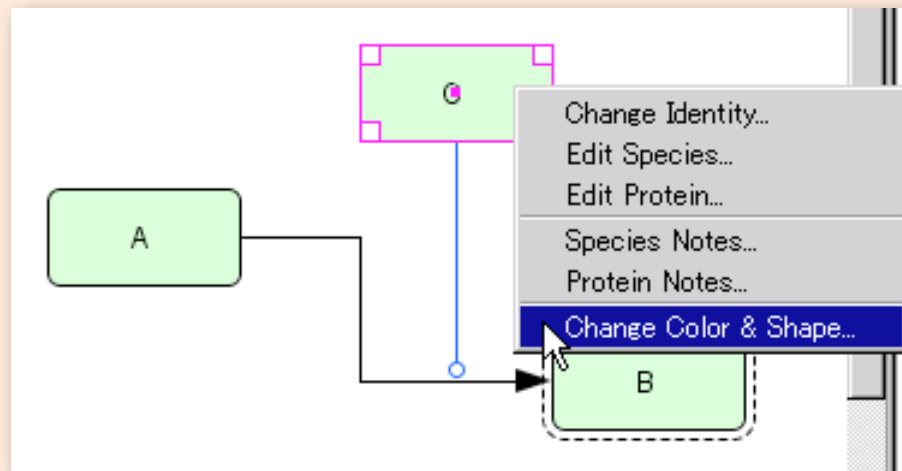
Set Active state

-
- Select Protein “B” [Component] → [Set Active]



Change Color

- Right click on Protein “C”
- Select [Change Color & Shape...]

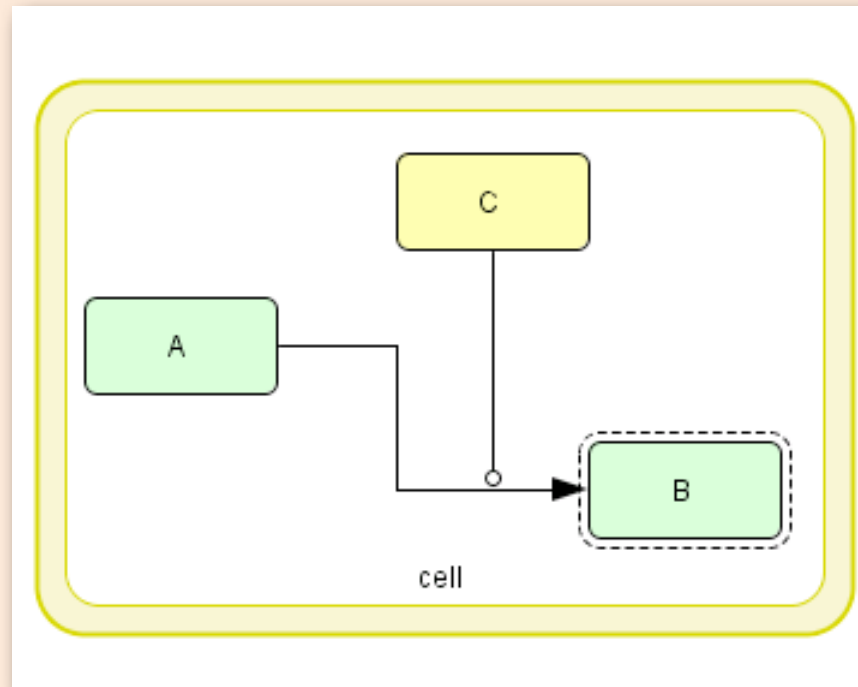


Compartment

- Click [Compartment] icon

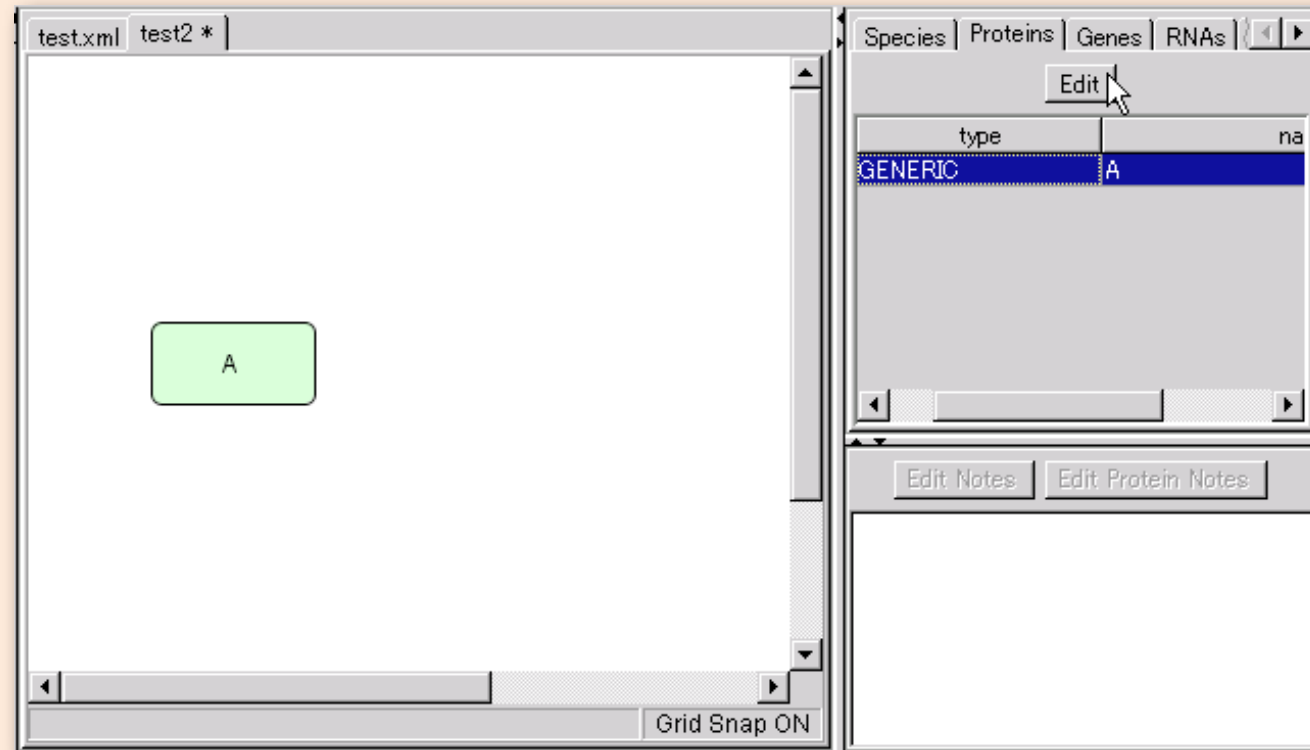


- Drag mouse cursor to specify its area
- Input name of compartment



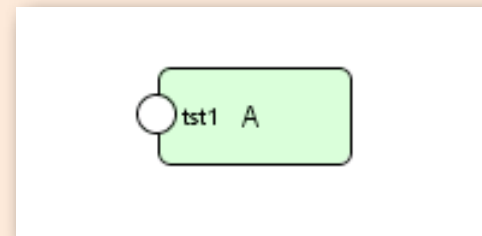
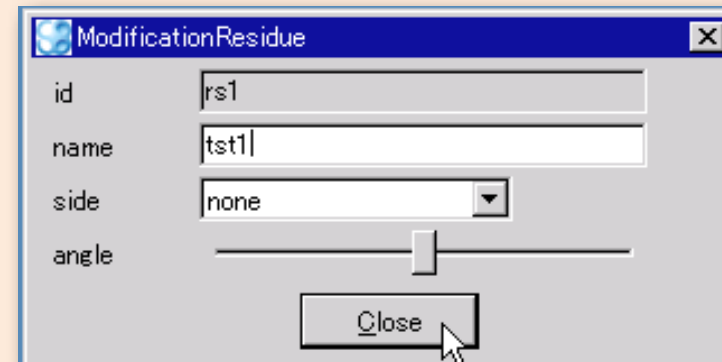
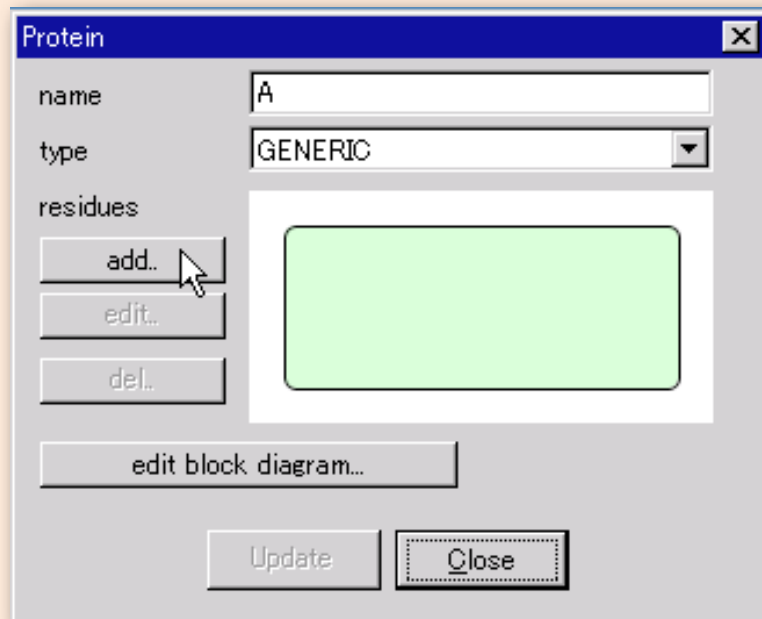
Add Residue to Protein

- Create new model (test2)
- Create Protein “A”
- Select Protein “A” in [Proteins] Tab Click
- [Edit] button



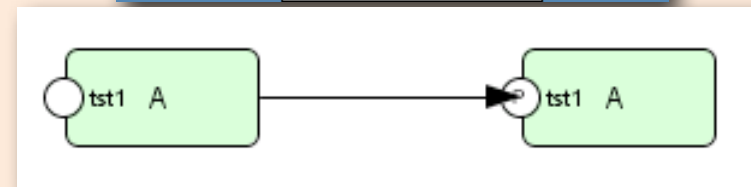
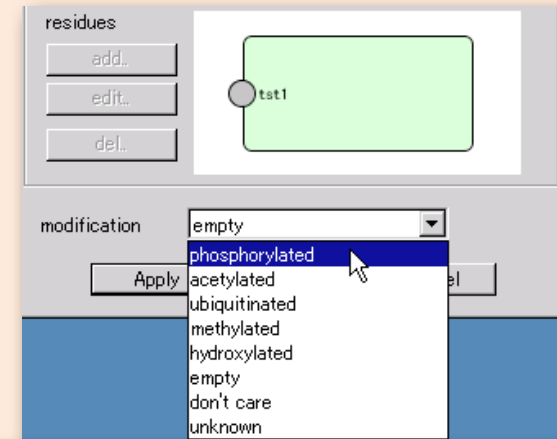
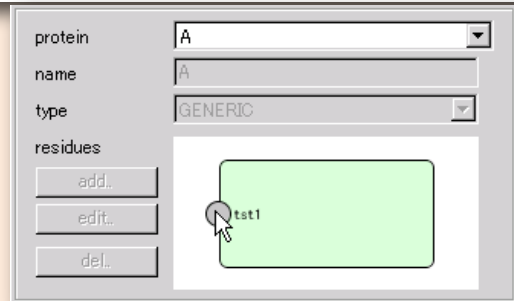
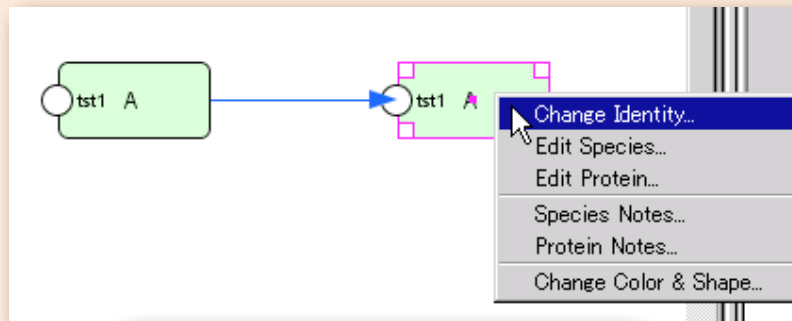
Add Residue to Protein

- Click [add] button on [Protein] dialog Input
- name for the residue (tst1) Click [Close]
- button
- Click [Update] Button



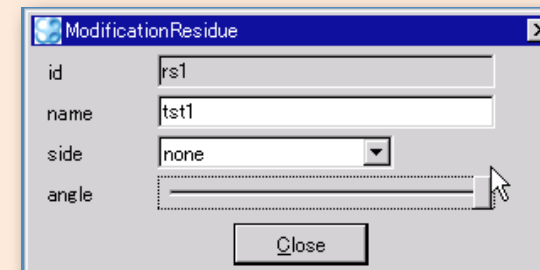
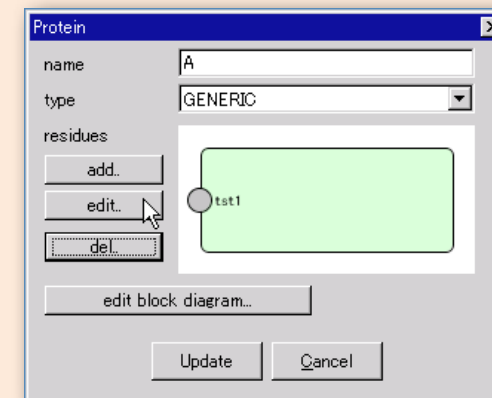
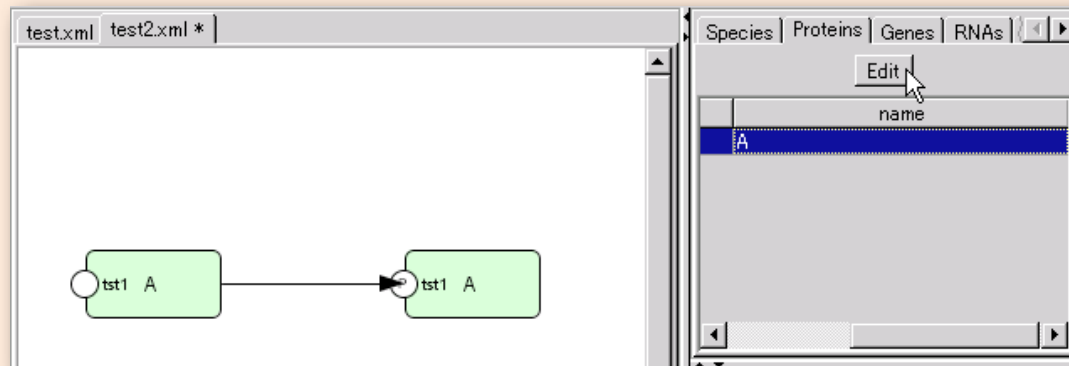
Add Residue to Protein

- Copy & Paste Protein “A” and then draw “State Transition” arrow
- Right Click on “A” (right side) and select [Change Identity...]
- Click residue “tst1” in Dialog
- Select [phosphorylated] in modification



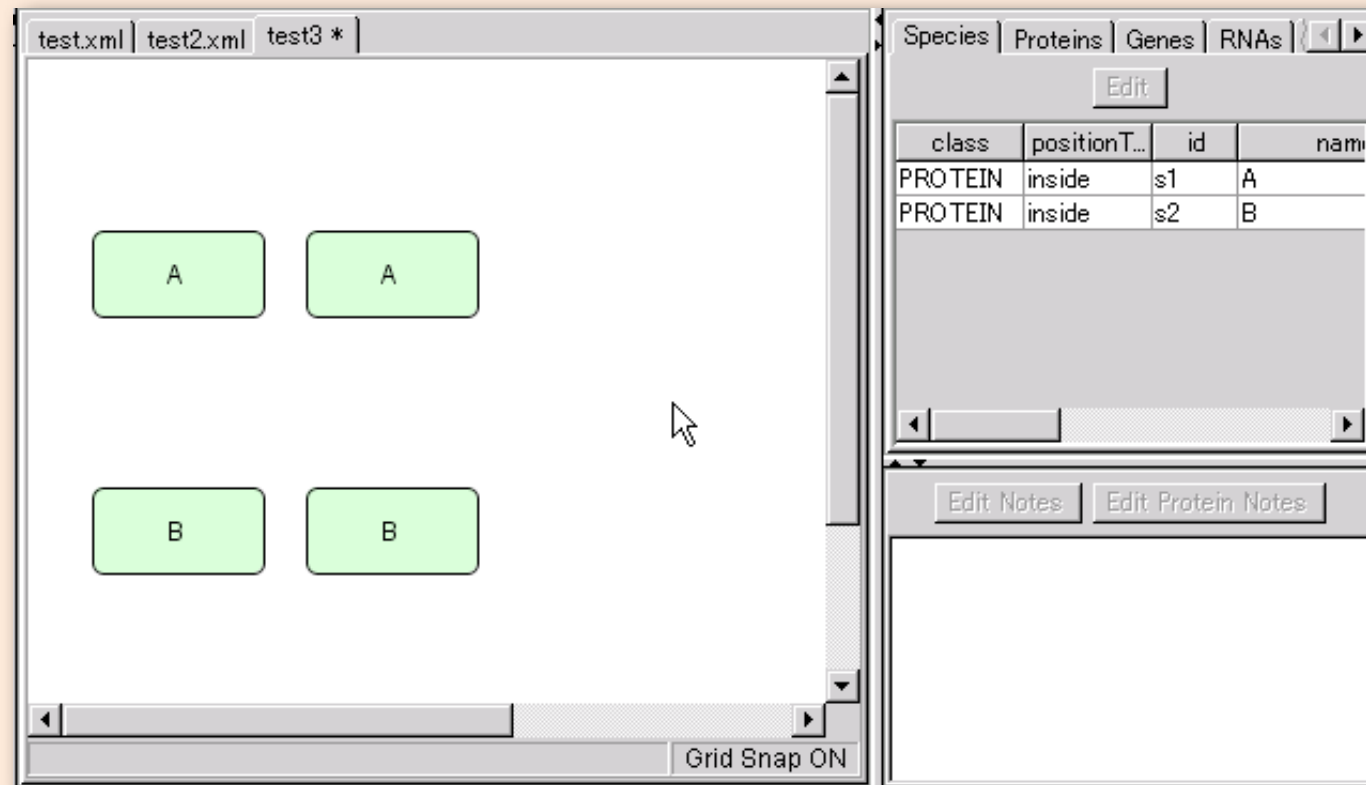
Change position of Residue

- Select Protein “A” in [Proteins] Tab Click
- [Edit] button
- Click residue “tst1” in Dialog Click
- [edit] button
- Drag [angle] sliderbar



Complex

- Create new model (test3)
- Create Proteins “A” and “B”
- Copy & Paste both “A” and “B”

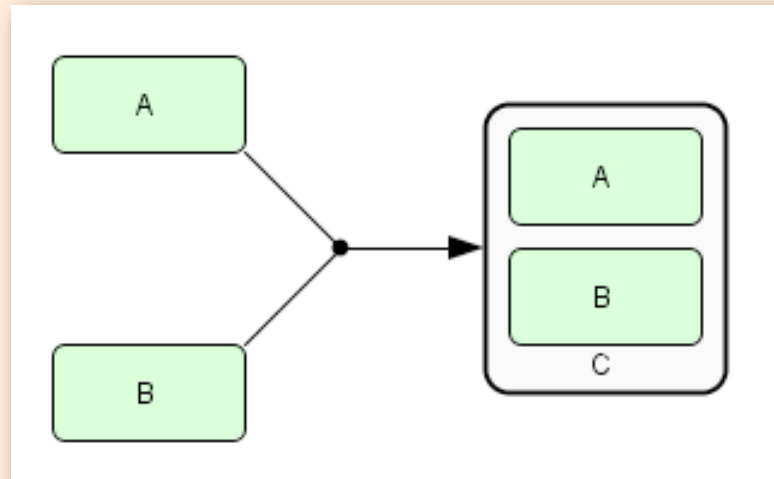


Complex

- Click [Complex] icon and create complex “C”

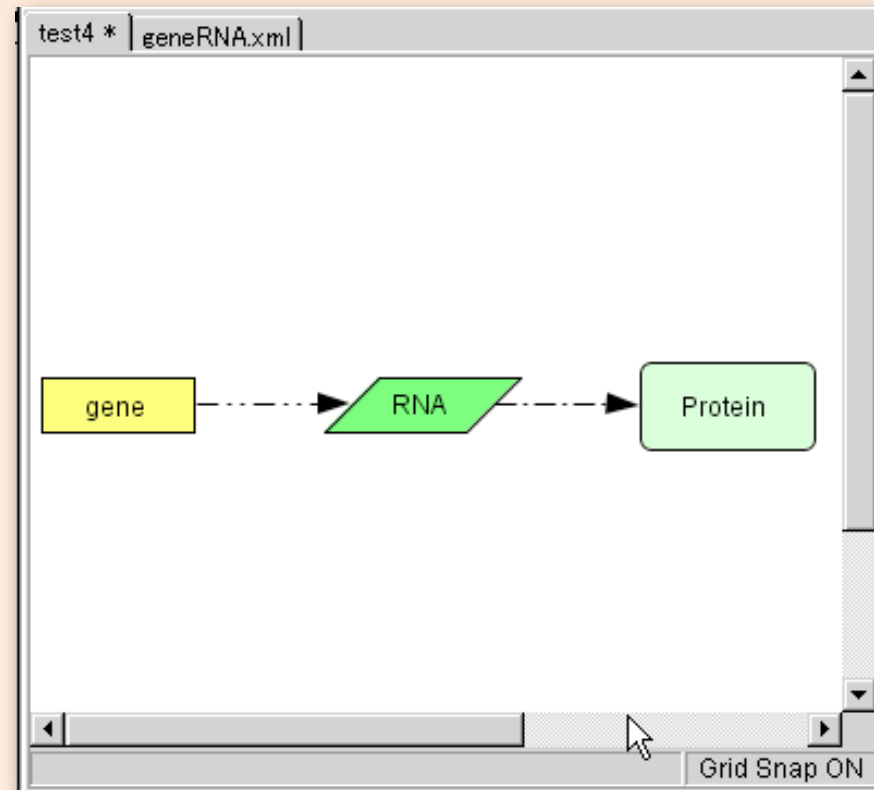


- Drag Protein “A” and “B” into complex C
- Draw “Association” arrow



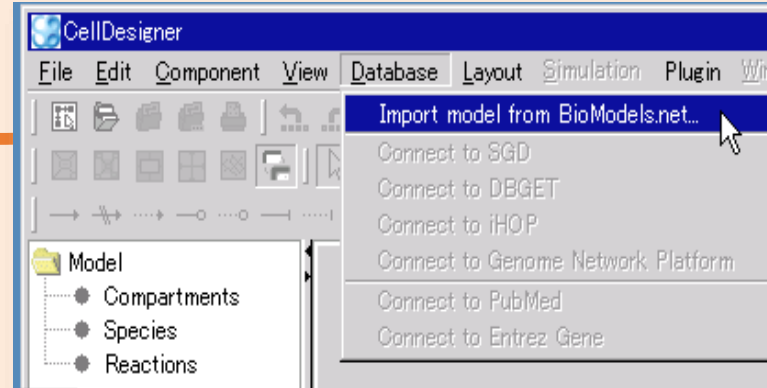
Gene & RNA

- Create new model (test4)
- Create gene, RNA and Protein
- Draw “Transcription” and “Translation”



Database Connection

Import model from
BioModels.net



The screenshot shows the BioModels.net dialog box, which is a window for selecting a model to import. It contains a table with two columns: 'ID' and 'Name'. The table lists 25 models, with the 10th model, 'Kholodenko2000 - Ultrasensitivity and negative feedback bring...', selected. At the bottom of the dialog, there are buttons for 'Description', 'Reference', 'Import', and 'Cancel'.

ID	Name
BIOMD0000000001	Edelstein1996 - EPSP ACh event
BIOMD0000000002	Edelstein1996 - EPSP ACh species
BIOMD0000000003	Goldbeter 1991 - Min Mit Oscil
BIOMD0000000004	Goldbeter 1991 - Min Mit Oscil, Expl Inact
BIOMD0000000005	Tyson1991 - Cell Cycle 6 var
BIOMD0000000006	Tyson1991 - Cell Cycle 2 var
BIOMD0000000007	Novak1997 - Cell Cycle
BIOMD0000000008	Gardner1998 - Cell Cycle Goldbeter
BIOMD0000000009	Huang1996 - Ultrasensitivity in MAPK cascade
BIOMD0000000010	Kholodenko2000 - Ultrasensitivity and negative feedback bring...
BIOMD0000000011	Levchenko2000_MAPK_noScaffold
BIOMD0000000012	Elowitz2000 - Repressilator
BIOMD0000000013	Poolman2004_CalvinCycle
BIOMD0000000014	Levchenko2000_MAPK_Scaffold
BIOMD0000000015	Curto1998 - purine metabolism
BIOMD0000000016	Goldbeter1995_CircClock
BIOMD0000000017	Hoefnagel2002_PyruvateBranches
BIOMD0000000018	Morrison1989 - Folate Cycle
BIOMD0000000019	Schoeberl2002 - EGF MAPK
BIOMD0000000020	hodgkin-huxley squid-axon 1952
BIOMD0000000021	Leloup1999_CircClock
BIOMD0000000022	Ueda2001_CircClock
BIOMD0000000023	Rohwer2001_Sucrose
BIOMD0000000024	Scheper1999_CircClock
BIOMD0000000025	Smolen2002_CircClock

- Compartments
- Species
- Reactions

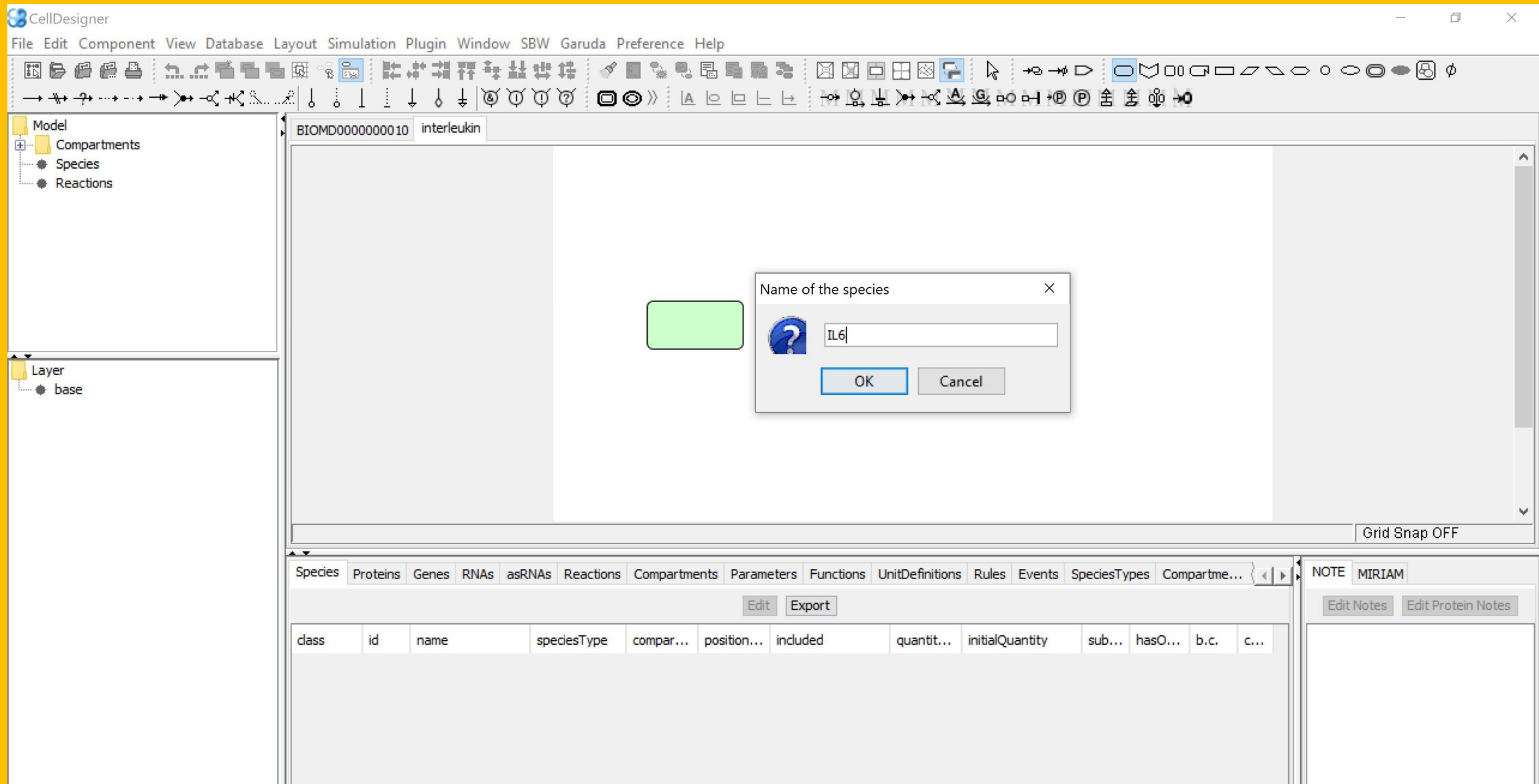
- base

Species	Proteins	Genes	RNAs	asRNAs	Reactions	Compartments	Parameters	Functions	UnitDefinitions	Rules	Events	SpeciesTypes	Compartment...
---------	----------	-------	------	--------	-----------	--------------	------------	-----------	-----------------	-------	--------	--------------	----------------

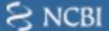
Edit Notes Edit Protein Notes

class	id	name	speciesType	compar...	position...	included	quantit...	initialQuantity	sub...	hasO...	b.c.	c...
PROTEIN	MKKK	Mos		uVol	transme...		Concentr...	90.0		false	false	false
PROTEIN	MKK...	Mos-P		uVol	transme...		Concentr...	10.0		false	false	false
PROTEIN	MKK	Mek1		uVol	transme...		Concentr...	280.0		false	false	false
PROTEIN	MKK_P	Mek1-P		uVol	transme...		Concentr...	10.0		false	false	false
PROTEIN	MKK...	Mek1-PP		uVol	transme...		Concentr...	10.0		false	false	false
PROTEIN	MAPK	Erk2		uVol	transme...		Concentr...	280.0		false	false	false

3



Database -> connect to Entrez Gene

 [Resources](#) [How To](#) [Sign in to NCBI](#)

Gene

[Create RSS](#) [Save search](#) [Advanced](#) [Help](#)

COVID-19 is an emerging, rapidly evolving situation.
Get the latest public health information from CDC: <https://www.coronavirus.gov>.
Get the latest research from NIH: <https://www.nih.gov/coronavirus>.
Find NCBI SARS-CoV-2 literature, sequence, and clinical content: <https://www.ncbi.nlm.nih.gov/sars-cov-2/>.

Gene sources
Genomic

Categories
Alternatively spliced
Annotated genes
Non-coding
Protein-coding

Sequence content
CCDS
Ensembl
RefSeq
RefSeqGene

Status
☒ Current
[Clear all](#)
[Show additional filters](#)

Tabular 20 per page Sort by Relevance Send to:

GENE

Was this helpful?

[IL6 – interleukin 6](#)

[Homo sapiens \(human\)](#)

Also known as: BSF-2, BSF2, CDF, HGF, HSF, IFN-beta-2, IFNB2, IL-6

GeneID: 3569

[RefSeq transcripts \(5\)](#) [RefSeq proteins \(5\)](#) [RefSeqGene \(1\)](#) [PubMed \(4,649\)](#)

Filters: [Manage Filters](#)

Results by taxon

Top Organisms [\[Tree\]](#)

Homo sapiens (258)
Mus musculus (86)
Rattus norvegicus (26)
Bos taurus (22)
Tupaia chinensis (9)
All other taxa (559)
[More...](#)

Find related data

Database:

Inventory of features

Open a new
model

Create a
reaction

Add anchor
point

Add catalytic
reaction

Set active
state

Change color

Include
compartment

Add residue to
protein

Change
position of
residue

Create
complex

Include genes
and RNAs

Connect to
databases

Choose layout

http://www.celldesigner.org/

Documents Help Model Repository Forum Plugins News Links Contacts

Help - Home > Quick Tutorial

Quick Tutorial

- Getting to Know CellDesigner
 - Starting the program
 - Navigation
- Creating a new Model
 - Species
 - Add a Protein on the Canvas
 - Add Protein Residues
 - Change the Residue/Region Status
 - Create Reaction
 - Tidy up your diagram layout
 - Export Images / Print Images
 - Customizing Properties
 - Add Notes (e.g. literature references) and MIRIAM to Proteins/Reactions
 - Refer to the databases
- Running the simulation
 - Set the Kinetic Laws / Parameters
 - Set the values of the Species
 - Using Control Panel for simulation
- Import / Export SBML file
- Viewing a Model
 - Connect to Database
 - View Notes

If you are the first-time user we would recommend you to **Guide Ver. 4.3**, which gives instruction using sample files.

Past Tutorial Presentations available:

- ICSB-2010 (Oct 10th, 2010)
- ICSB 2009
- Tutorial for Simulation (Adams)

CellDesigner.org

ENHANCED BY Google

home | features | downloads | documents | models | forum | news | links | contacts

CellDesigner™: A modeling tool of biochemical networks

Download CellDesigner **Current Release Version: CellDesigner 4.4.2**
macOS Catalina and Ubuntu 18.04 support + Plugin APIs enhances + BioModels new API support + Garuda enabled + bug fixes. find out more...

You do not have to install JVM separately as it is included in the installer.

Check also:

- CellDesigner on Garuda platform (Ver4.4.1 Win / Mac)
- Plugins / Utilities
 - CellDesigner Plugin API Document (ver4.4.1)
- Models built with CellDesigner
- BioModels.net models simulation results with CellDesigner 4.0

What is CellDesigner™

CellDesigner™
Ver. 4.3

© 2003 - 2013 The Systems Biology Institute. All rights reserved.
http://celldesigner.org

CellDesigner - 4.3.0 - Tue., 29 Feb. 2013

Headlines

CellDesigner 4.4.2 Mac installer updated for Catalina Support (2020/3/30)

CellDesigner 4.4.2 is now available (2019/05/20)

CellDesigner on Garuda platform (2017/2/14) Garuda enabled Ver4.4 is available as Ver4.4.1 Win / Mac.

PhysioDesigner 1.0 is available. PhysioDesigner can embed CellDesigner's SBML model into its multilevel physiological PHML model. (2014/08/01)

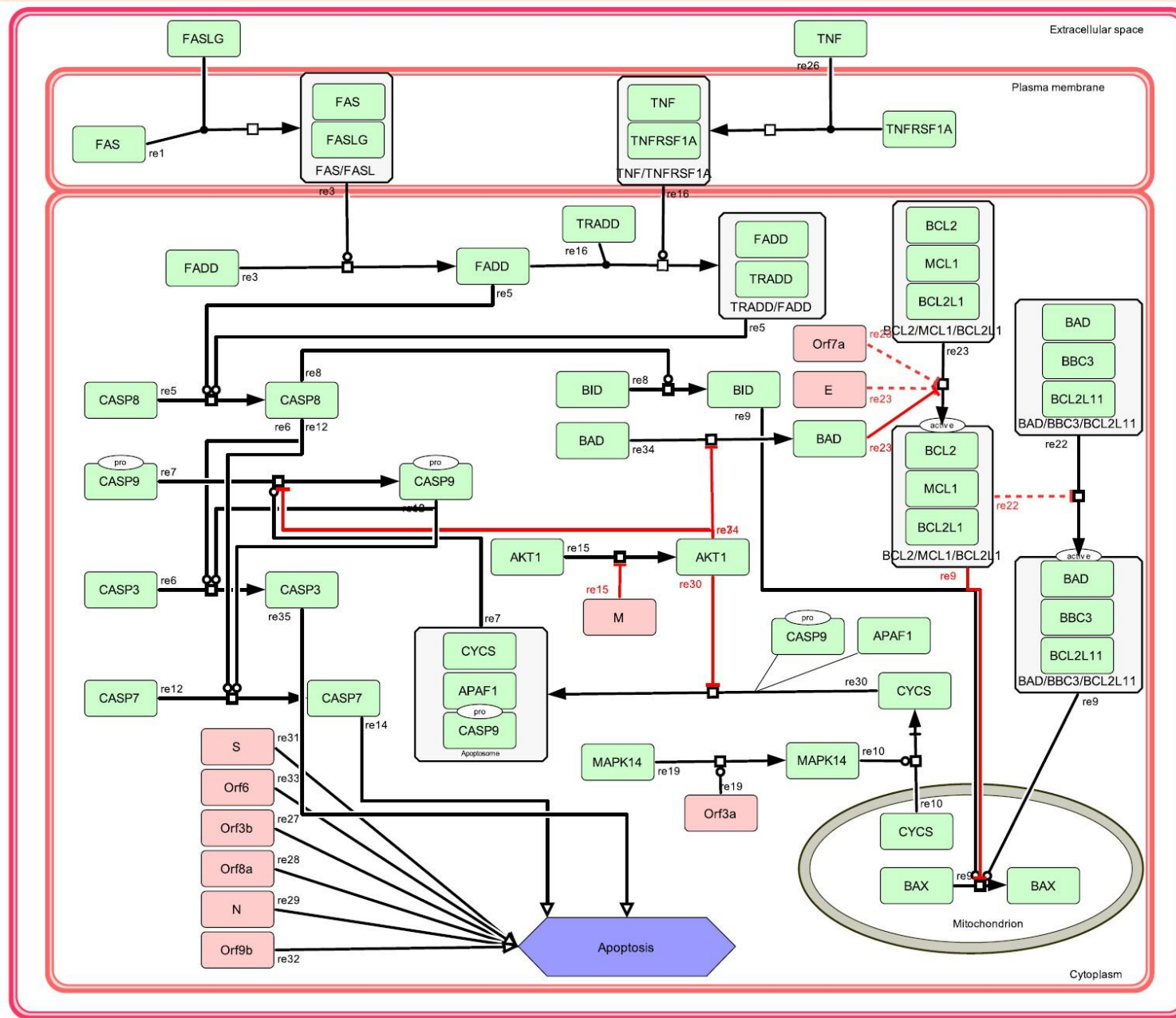
CellDesigner 4.4 is now available (2014/07/12)

Advertisement

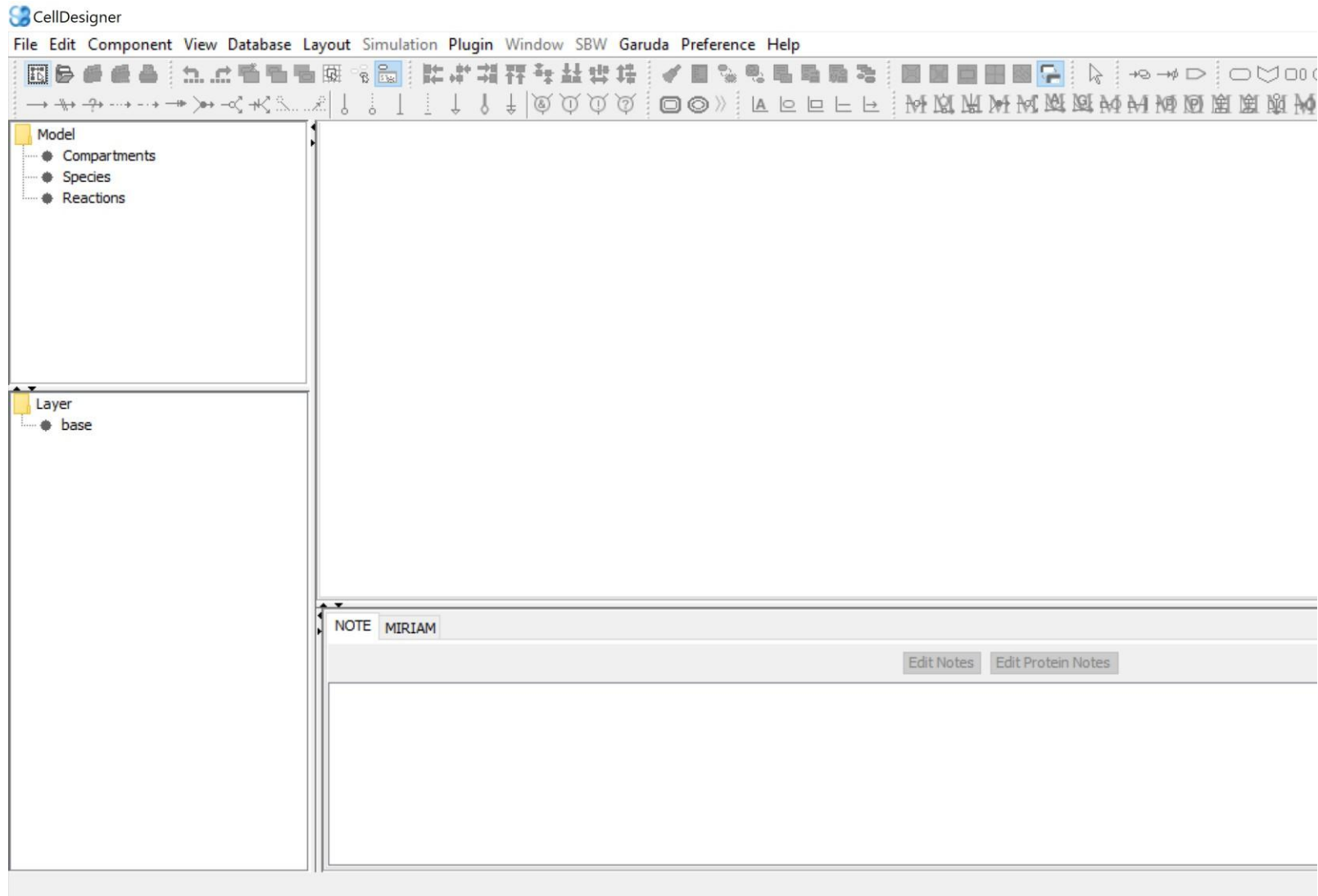
npj | Systems Biology and Applications

Launching in 2015

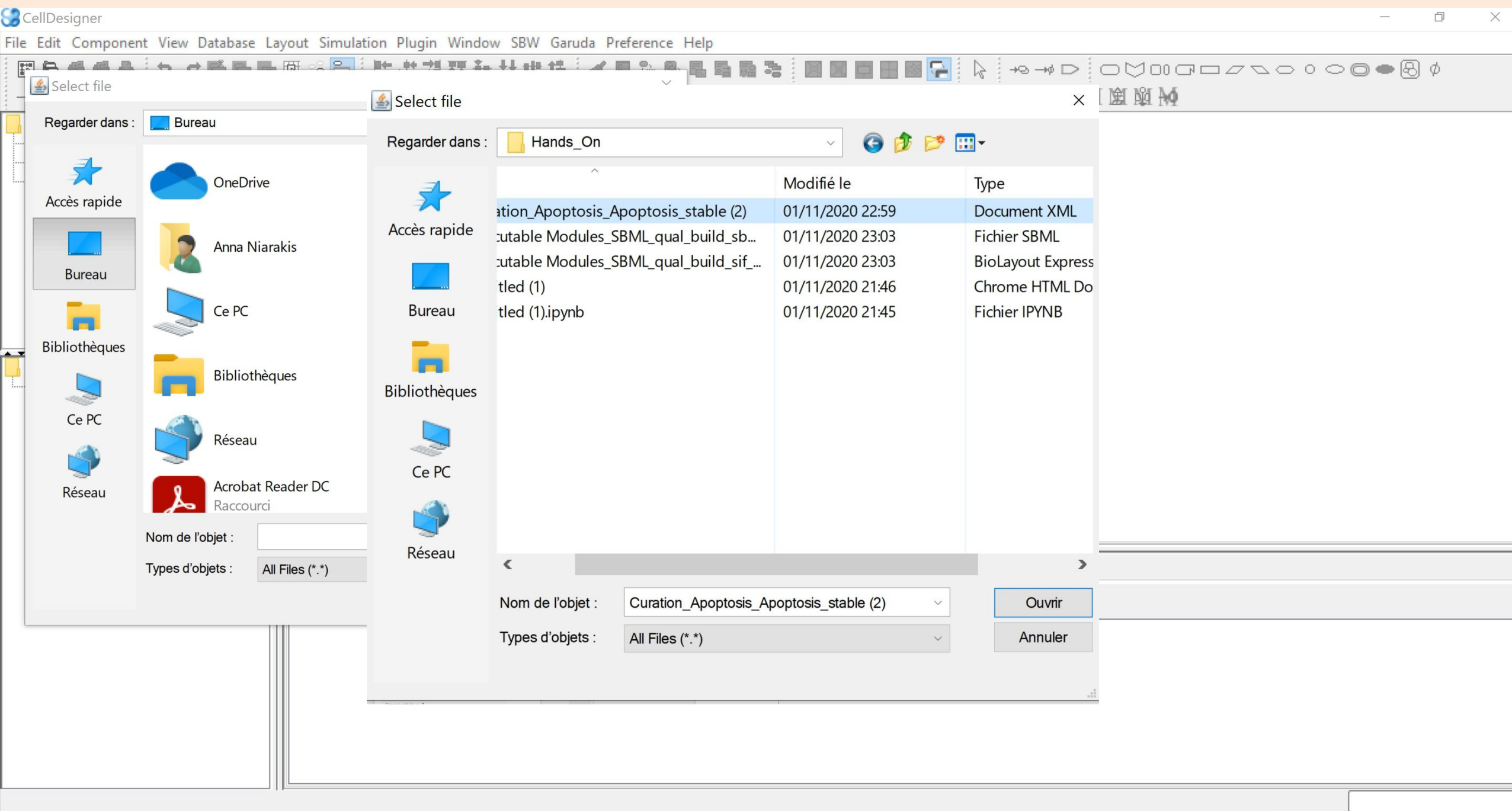
EDITOR-IN-CHIEF
Hiroaki Kitano

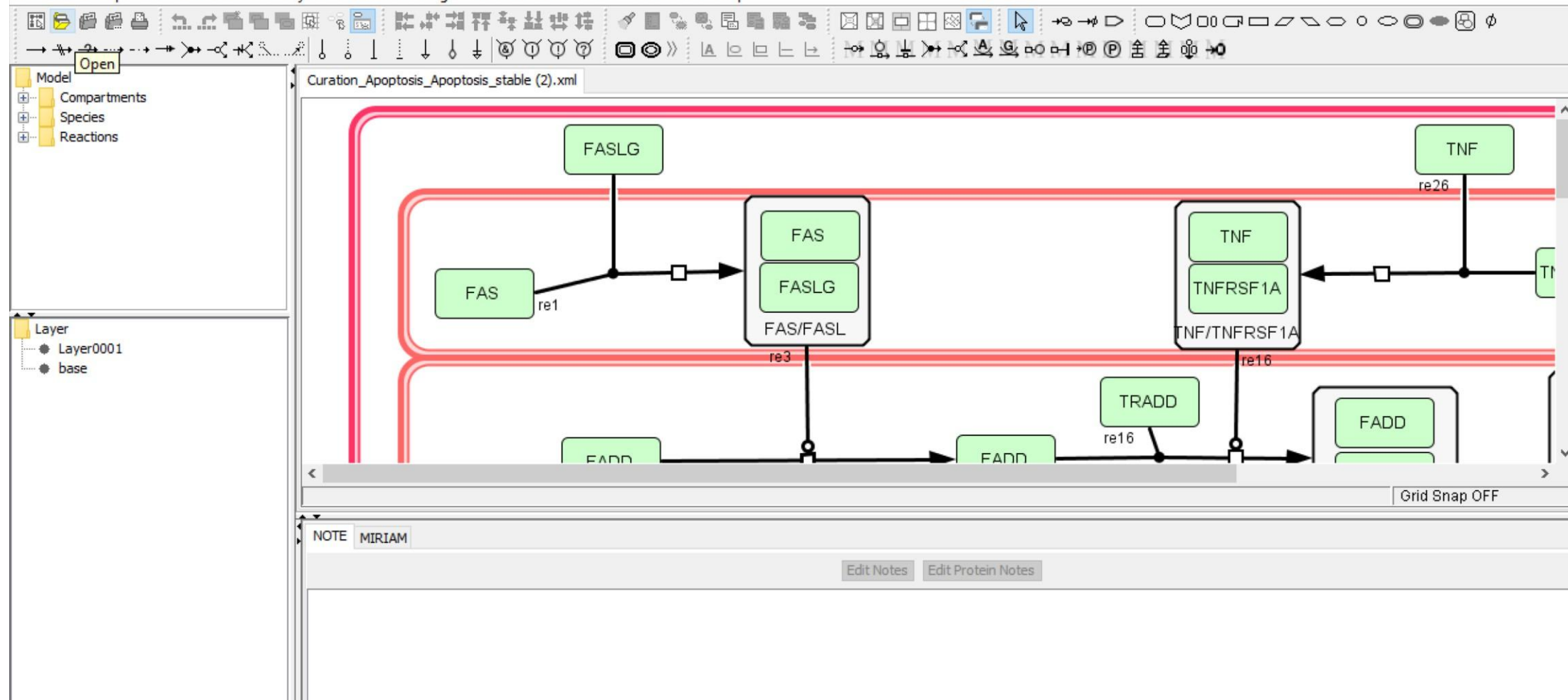


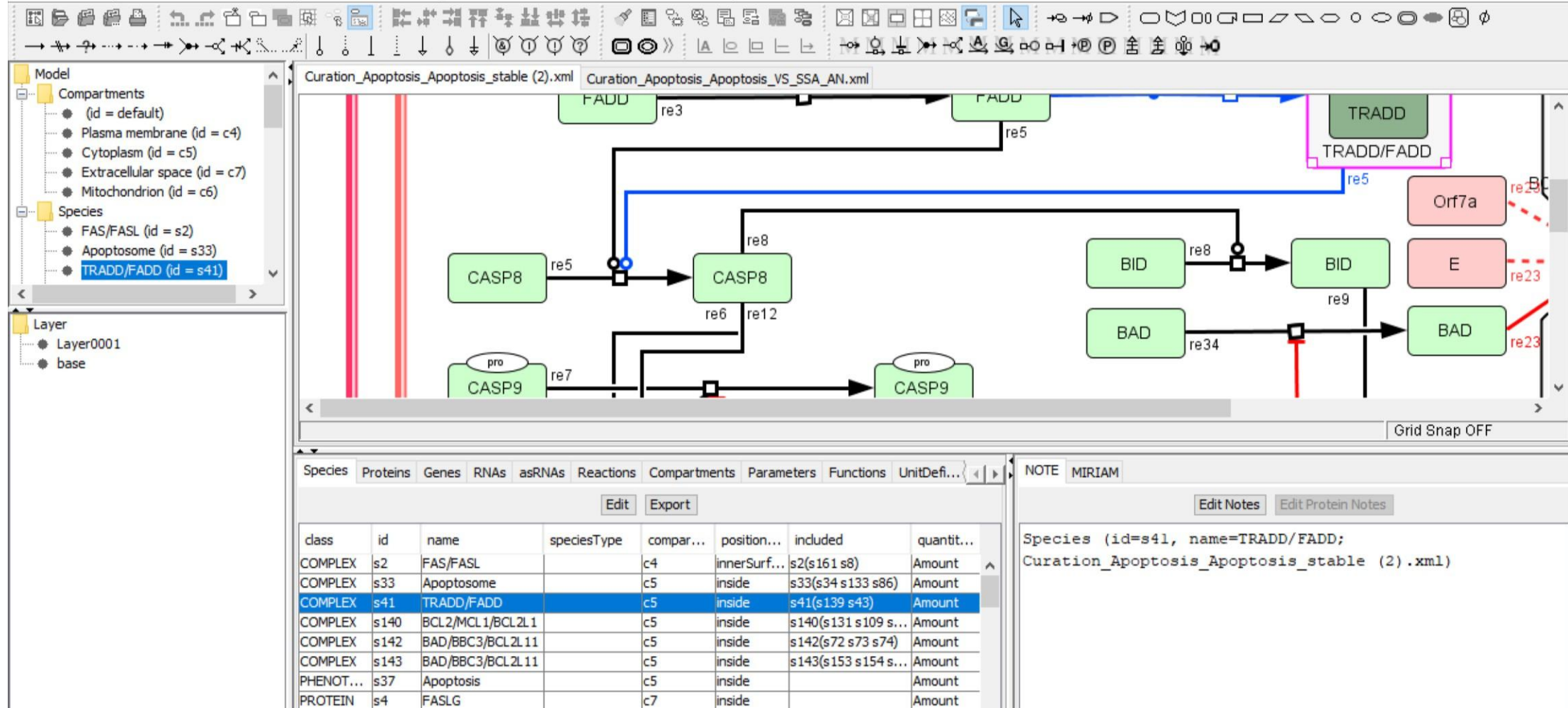
Open the Apoptosis xml diagram in CellDesigner

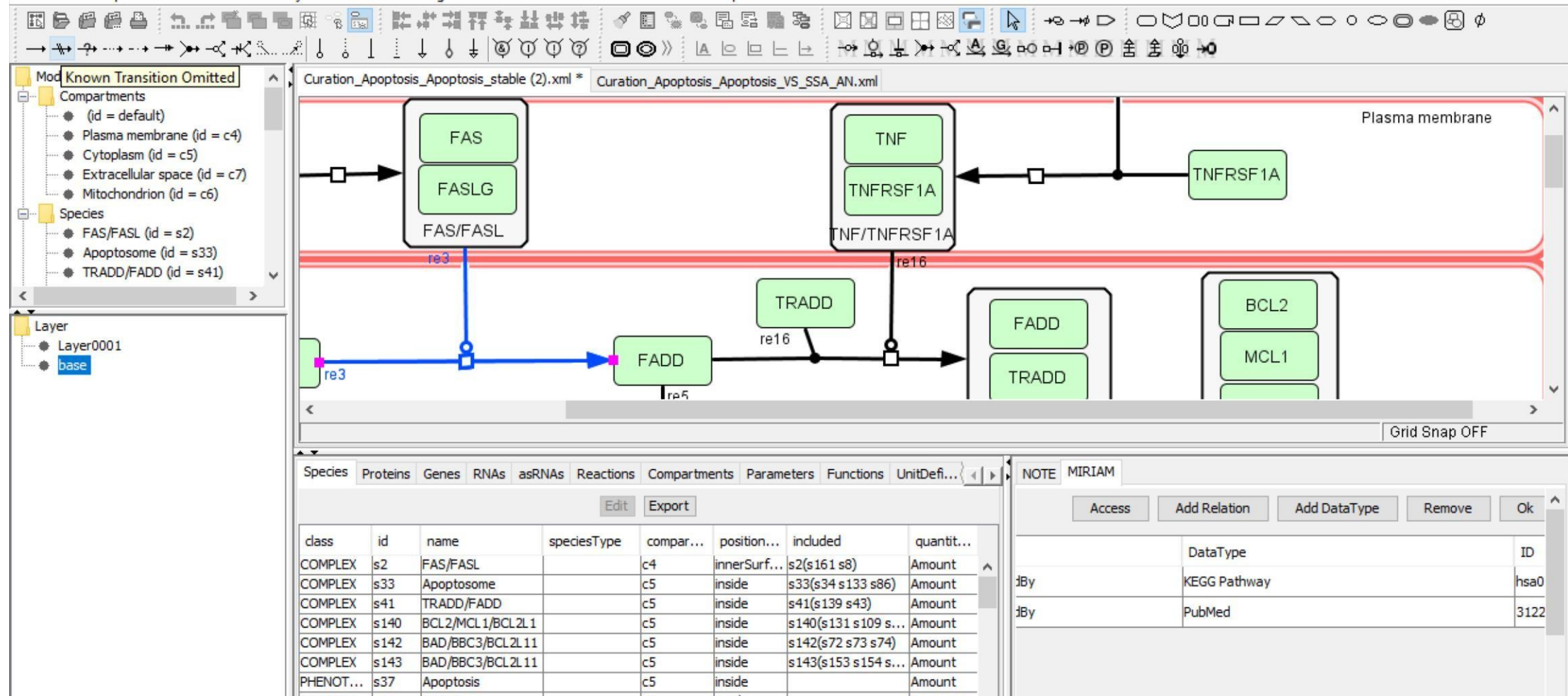


Open in
CellDesigner











```
1  <?xml version="1.0" encoding="UTF-8"?>
2  <sbml xmlns="http://www.sbml.org/sbml/level2/version4" xmlns:celldesigner="http://www.sbml.org/2001/ns/celldesigner" level="2" version="4">
3  <model metaid="Covid_19_apoptosis_21_04_2020" id="Covid_19_apoptosis_21_04_2020">
4  <notes>
5  <html xmlns="http://www.w3.org/1999/xhtml">
6  <head>
7  <title/>
8  </head>
9  <body>Vidisha Singh, GenHotel Laboratory, University of Evry-Paris Saclay, Evry, France
10 Apoptosis pathway for Covid-19 maps project is constructed by using the review paper of Fung&Liu,2019,Annu rev micro (PMID=31226023).
11 KEGG pathway database is used to add reactions between the elements.
12 Map is constructed in the form of a cell including plasma membrane, cytoplasm in addition to extracellular space.
13 Ligands are present in the extracellular space which interacts with receptors in the plasma membrane to initiate signaling cascade in the cytoplasm.
14 All the map elements are named according to HGNC naming scheme.
15 All map reactions are annotated with PubMed Ids, KEGG pathway Ids using MIRIAM add relation&bqbiol:isDescribedby&.
16 MESH Id is provided for the phenotype &Apoptosis&.
17 </body>
18 </html>
19 </notes>
20 <annotation>
21 <celldesigner:extension>
22 <celldesigner:modelVersion>4.0</celldesigner:modelVersion>
23 <celldesigner:modelDisplay sizeX="1350" sizeY="1200"/>
24 <celldesigner:listOfIncludedSpecies>
25 <celldesigner:species id="s8" name="FASLG">
26 <celldesigner:notes>
```